

A Model Confidence Set for Learner Selection in Generic Machine Learning

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Abstract

Generic Machine Learning (Chernozhukov et al., 2025a) provides valid inference on heterogeneous treatment effects using any machine learning algorithm to proxy the conditional average treatment effect, but offers limited guidance on which learner’s results to report when multiple learners are employed. Existing reporting practice typically defaults to ad hoc selection of a top-performing learner or to a Bonferroni correction across all candidates, neither of which adapts to the relative performance of the learners in the data. I propose applying the model confidence set of Hansen et al. (2011) to the split-level goodness-of-fit values produced by GML, yielding a data-driven set of learners whose performances are statistically indistinguishable, a set that collapses to a singleton when one learner clearly dominates. Inference then proceeds over this set rather than the full collection of candidates. An initial split of the sample reserves a portion of the data exclusively for learner selection, so that the reported inference uses only observations that never entered the selection stage. The resulting two-stage procedure carries an overall coverage guarantee of $1 - \alpha_1 - \alpha_2$, with separate budgets for selection and inference error. Monte Carlo simulations show that the procedure controls size, sacrifices little power relative to naive selection, and reveals variation in point estimates across surviving learners when the data do not support a unique choice. In an application to the lending experiment of Bryan et al. (2024), formal learner selection recovers a significant boost in profits for the top quartile of firms at the conventional 90% level that is otherwise lost to multiplicity correction across all candidate learners. A second application to the groundwater conservation experiment of Hrozencik et al. (2024) illustrates a complementary case: the MCS identifies a relatively superior learner, while subsequent GML inference provides little evidence that even the selected proxy captures systematic treatment-effect heterogeneity. Together, the applications show that formal learner selection can sharpen inference when informative proxies exist without manufacturing evidence of heterogeneity when they do not.

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1 Introduction

Generic Machine Learning (Chernozhukov et al., 2025a) provides a powerful framework for inference on heterogeneous treatment effects in randomized experiments. The method is deliberately agnostic with respect to the machine learning algorithm used to proxy the conditional average treatment effect (CATE): any learner, or collection of learners, may be employed in the first stage, and valid inference on key parameters follows from sample splitting and median aggregation in the second stage. This agnosticism is a central feature of the framework. Researchers rarely have an *a priori* basis for choosing which learning algorithm will best detect the underlying patterns of treatment effect heterogeneity in a given dataset, and the candidates available—random forests, lasso, neural networks, causal forests, and many others—embody substantially different inductive biases.

In practice, however, most applied work using GML has relied on a small, recurring set of four to five learners, commonly random forests, gradient boosting, regularized regression, and neural networks. This convention is understandable: with a small learner set, one can apply a Bonferroni correction across all learners and still obtain reasonably tight confidence intervals. But this approach is at odds with the agnostic philosophy of GML. A richer set of candidate learners increases the probability that at least one learner captures the relevant structure in the true CATE, particularly when the functional form is complex or involves interactions among covariates. The cost of expanding the learner set under current practice is a loss of statistical power, since a Bonferroni correction over L learners allocates the error rate across all L candidates and therefore requires increasingly conservative critical values. As L grows, the inferential gains from a better proxy may consequently be offset by wider confidence intervals.

This tension between exploration and power gives rise to a learner selection problem. If the researcher could identify a small subset of learners whose performance is statistically indistinguishable from the best, she could apply a multiplicity correction only over this subset, preserving power while maintaining the benefits of a large initial candidate set. The challenge is to construct such a subset in a principled, data-driven manner. Chernozhukov et al. (2025a) suggest reporting the union of confidence intervals or applying a Bonferroni correction when two learners are “near-winners,” but leave open the question of how to determine which learners qualify as near-winners and how to proceed when more than two are competitive.

This paper proposes a solution based on the model confidence set (MCS) of Hansen et al. (2011). The MCS is a sequential testing procedure that begins with the full set of candidate learners and eliminates those whose goodness-of-fit is significantly inferior, terminating when the surviving learners cannot be statistically distinguished from one another. The resulting set contains the best learner with a user-specified probability, analogous to a confidence interval for a parameter. When applied to GML, the MCS provides a formal answer to the near-winner question: the near-

winner set is precisely the model confidence set, and its cardinality adapts to how informative the data are about relative learner performance. In settings where a single learner clearly dominates its competitors, the MCS shrinks to a singleton and no multiplicity correction across surviving learners is needed. When several learners perform comparably, the set expands and the correction is applied only over the relevant competitors.

I illustrate the procedure in two randomized experiments that highlight complementary features of the learner-selection problem. The first is the loan experiment of Bryan et al. (2024), which is an especially useful setting because the authors implement repeated-split GML with four competing learners and take a conservative approach to reporting confidence intervals by applying a Bonferroni adjustment to cover all four learners. Using the corrected split-level results following the published corrigendum, I show that the MCS selects a singleton learner and that formal selection changes the substantive inference: the positive profit effect for the top quartile of predicted beneficiaries remains statistically significant at the conventional 90% level, whereas it does not survive a proper Bonferroni correction across all four initially considered learners. The second application uses the groundwater conservation experiment of Hrozencik et al. (2024). In this setting, the MCS again identifies a relatively superior learner, but subsequent GML inference provides little statistically detectable evidence that even the selected proxy is aligned with heterogeneous treatment effects. The two applications therefore emphasize an important distinction: selecting the best-performing learner among a candidate set is not equivalent to establishing that the selected learner is a strong proxy for the CATE. Formal learner selection can sharpen inference when informative heterogeneity proxies exist without manufacturing evidence of heterogeneity when they do not.

I establish the theoretical validity of this two-stage procedure—MCS for learner selection followed by union-based inference over the selected set—using an initial split of the sample that separates the observations used for selection from those used for the reported inference. The overall coverage probability is bounded below by $1 - \alpha_1 - \alpha_2$, where α_1 governs the MCS and α_2 governs the inference stage. Monte Carlo simulations using a benchmark DGP from the recent discussion of GML (Wager, 2025; Chernozhukov et al., 2025b) demonstrate that the procedure controls size, preserves power relative to naive selection, and provides information about which learners are empirically competitive under the chosen GML performance criterion. Beyond hypothesis testing, the MCS offers a practical diagnostic: the range of point estimates across the surviving learners quantifies the sensitivity of the reported magnitude to the choice of learner, information that is unavailable under naive top-1 selection.

1.1 Related Literature

This paper contributes to several strands of the literature on post-selection inference, multiple comparison procedures, and machine learning for causal inference.

The methodological foundation is Generic Machine Learning (Chernozhukov et al., 2025a), which provides a framework for inference on heterogeneous treatment effects that is agnostic to the choice of ML learner. Chernozhukov et al. (2025a) acknowledge that the question of which learner to report remains open, suggesting either the union of confidence intervals or a Bonferroni correction when two learners are close in performance. Wager (2025) explores whether single-split and sequential procedures can reduce the computational burden of repeated splitting, while Imai and Li (2025) propose a cross-fitting approach based on randomization inference that requires only a single partition of the data. In their rejoinder, Chernozhukov et al. (2025b) demonstrate that the multi-split approach offers lower estimation risk and greater robustness to data mining relative to single-split alternatives. The present paper takes the multi-split framework as given and addresses the complementary problem of selecting among learners within it.

The learner selection problem in GML can be viewed as an instance of inference after selection, a topic that has received substantial attention in the statistics literature. Berk et al. (2013) develop valid post-selection inference for linear models, providing confidence intervals that account for all possible model selection procedures applied to the data. Fithian et al. (2017) characterize optimal inference after model selection by conditioning on the selection event, establishing a general framework based on sufficient statistics and the selection-adjusted likelihood. Andrews et al. (2024) study inference on the “winner” among competing estimates, deriving confidence intervals for the parameter associated with the largest observed value. Their setting is closely related to the GML learner selection problem: the researcher observes multiple estimates (one per learner), selects the best, and wishes to perform valid inference on the selected parameter. Zrnic and Fithian (2024) propose the zoom correction, a flexible method for inference on the winner that adapts to the gap between the best and remaining candidates, yielding tighter intervals when a clear winner emerges. I explored the zoom correction in the GML context but found that the power loss in low-signal settings was substantial, and the single-split requirement sacrificed the robustness benefits of median aggregation. More broadly, the sample-splitting approach adopted here is a standard device for post-selection inference: by using disjoint observations for the selection and inference stages, one can appeal to the point-wise validity of the underlying GML machinery without the technical complications of conditioning on the selection event.

The model confidence set of Hansen et al. (2011) provides a different approach to selection. Rather than correcting inference for a selected model, the MCS constructs a set of models whose performances are statistically indistinguishable, in the same spirit as a confidence interval for a parameter. The MCS has been widely applied in forecasting and volatility model comparison (see, e.g., Hansen et al., 2011, and references therein), but has not previously been applied to learner selection in causal inference settings. The multiple testing procedures of Romano and Wolf (2007) offer a related framework for controlling familywise error rates across simultaneous hypotheses using resampling methods. While a Romano-Wolf stepdown procedure could in principle be applied to

the GML learner comparison problem, doing so requires estimating the joint covariance of median-aggregated test statistics across learners, which presents difficulties when influence functions are only valid conditional on their respective sample splits. The MCS sidesteps this issue by operating directly on split-level performance values.

2 Theory

The main focus of this paper is an application of the model confidence set (MCS) to learner results in Generic Machine Learning (GML) of Chernozhukov et al. (2025a). Hansen et al. (2011) does not make any assumptions about a model’s relationship to the underlying DGP, rather, the model confidence set relies on a user-specified loss criterion of sample performance. The MCS procedure begins with a (large) set of candidate models, and removes candidates that are “not as good”—based on an equivalence test and an elimination rule—until the set of models is pruned to a set of models whose sample performances cannot be distinguished from one another for a given significance level. MCS has been applied in many ML settings involving time series data and forecasting (e.g., (Goulet Coulombe et al., 2022)), but has not been a popular approach for learner performance in causal inference settings. Although MCS was developed in the context of time series models, in this section I show that the MCS procedure—with a few tweaks—can be applied to the GML learner-selection problem.

2.1 Generic Machine Learning

GML offers a unique approach to estimating key features of the CATE, which happens in two stages which we might call *learning* and *post-processing*. Before learning, data $\mathcal{D} = (Y_i, D_i, Z_i)_{i=1}^N$, where Y_i is the outcome, D_i is a binary treatment indicator, and Z_i is a vector of covariates, are partitioned into an auxiliary sample $\mathcal{D}_A$ and a main sample $\mathcal{D}_M$. It may be more convenient for some readers to think of $\mathcal{D}_A$ and $\mathcal{D}_M$ as training and evaluation samples, respectively. Following Chernozhukov et al. (2025a), the partitions are assumed to be of roughly equal size ($N/2$).

Stage 1 (Learning): Let $Y(1)$ denote the outcome under the treated state and $Y(0)$ under non-treated. Using $\mathcal{D}_A$, estimate the baseline conditional average $b_0(Z) = \mathbb{E}[Y(0) \mid Z]$ and CATE $\mathbb{E}[Y(1) - Y(0) \mid Z]$ using ML methods (i.e., learners) $\ell = 1, \dots, L$. Let the learned *proxy predictors* for the BCA and CATE (using $\mathcal{D}_A$) be denoted $B(Z)$ and $S(Z)$, respectively.

Stage 2 (Post processing): I consider two inferential targets in the post-processing stage for

the MCS procedure. First, the best linear predictor (BLP) of $s_0(Z)$ by the proxy $S(Z)$:

$$BLP[s_0(Z) | S(Z)] = \mathbb{E}[s_0(Z)] + \frac{\text{Cov}[s_0(Z), S(Z)]}{\text{Var}[S(Z)]}(S(Z) - \mathbb{E}[S(Z)]) \quad (1)$$

$$= \beta_1 + \beta_2 (S(Z) - \mathbb{E}[S(Z)]). \quad (2)$$

The BLP is identified by the weighted residual linear projection (WR BLP)

$$Y = \alpha'_0 X_1 + \beta_1(D - p(Z)) + \beta_2(D - p(Z))(S(Z) - \mathbb{E}[S(Z)]) + \epsilon, \quad \mathbb{E}[w(Z)\epsilon X] = 0, \quad (3)$$

where $p(Z) = \Pr(D = 1 | Z)$ is the propensity score, $w(Z) = \{p(Z)(1 - p(Z))\}^{-1}$ are the weights, and X_1 includes a constant, $B(Z)$ and other baseline controls. In a randomized experiment with constant propensity p , the weights simplify to a constant. The parameter β_2 is of primary interest: $\beta_2 = 1$ indicates that $S(Z)$ perfectly proxies the CATE, $\beta_2 = 0$ indicates that $S(Z)$ is uncorrelated with $s_0(Z)$ (or that $s_0(Z)$ is constant), and intermediate values reflect partial proxy quality. Note that if no heterogeneity is present in the true CATE, then β_2 is zero. Thus, a hypothesis test of $\beta_2 = 0$ is often utilized in practice as a test against zero heterogeneity. Using $\mathcal{D}_M$, one estimates $\hat{\beta}_1, \hat{\beta}_2$ and associated standard errors via weighted residual regression on (3) with $S(Z)$ replaced by $\hat{S}(Z)$ from the learning stage.

Second, sorted group average treatment effects (GATES). Partition observations in $\mathcal{D}_M$ into G groups based on the ranking of the proxy $\hat{S}(Z)$. In the simplest case with $G = 2$, let $G_k = \mathbf{1}\{\hat{S}(Z) \in \text{quantile group } k\}$ for $k = 1, 2$. The GATES are the group-specific average treatment effects $\gamma_k = \mathbb{E}[Y(1) - Y(0) | G_k = 1]$, estimated via the regression

$$Y = \alpha'_0 X_1 + \sum_{k=1}^G \gamma_k (D - p(Z)) G_k + \nu, \quad \mathbb{E}[w(Z)\nu X] = 0. \quad (4)$$

The contrast $\gamma_G - \gamma_1$ tests whether the group predicted to benefit most from treatment actually has a larger treatment effect than the group predicted to benefit least. As with the BLP, estimation proceeds by WR regression on $\mathcal{D}_M$ with learned proxies from $\mathcal{D}_A$.

Chernozhukov et al. (2025a) emphasize that their methodology is generic with respect to the choice of machine learners used to form proxy predictors, and that $B(Z)$ and $S(Z)$ are generally noisy and inconsistent (without imposing untestable assumptions).¹ They navigate around the issue of potentially-biased and noisy predictors via repeated sample splitting. Independently generated partitions $w = 1, \dots, W$ yield split-specific estimates $\hat{\theta}_{\ell w}$ and standard errors $\hat{\text{se}}_{\ell w}$ for each learner ℓ and target parameter θ (e.g., β_2 or the contrast $\gamma_G - \gamma_1$). Within each split, inference conditional on the auxiliary sample is approximately Gaussian by Lemma D.1 of Chernozhukov et al. (2025a).

¹See section 2.2-2.4, and the footnotes therein, of Chernozhukov et al. (2025a) for further explanation and a review of the issues faced in the ML + causal inference literature.

Across splits, point estimates and confidence bounds are median-aggregated, yielding valid unconditional inference that is robust to the randomness of any single partition. Specifically, within-split confidence intervals at nominal level ζ produce median-aggregated intervals with coverage at least $1 - 2\zeta$ (Theorems 4.2–4.3 of Chernozhukov et al. (2025a)). This within-learner guarantee is a per-learner statement and does not, on its own, control coverage after a data-dependent learner is selected; the two-stage procedure developed in Section 3 addresses this.

2.2 The Learner Selection Problem

Each learner ℓ produces a different proxy $\hat{S}_\ell(Z)$, and consequently different estimates of the inferential targets $\hat{\theta}_\ell$. To compare learners, Chernozhukov et al. (2025a) introduce the goodness-of-fit functional

$$\Lambda(\ell) = |\beta_{2,\ell}|^2 \text{Var}(S_\ell(Z)), \quad (5)$$

which can be interpreted as a measure of explained variation in the CATE attributable to proxy ℓ . Its sample analog, computed within each split, is $\hat{\Lambda}_w(\ell) = |\hat{\beta}_{2,\ell w}|^2 \text{Var}[\hat{S}_{\ell w}(Z) \mid \mathcal{D}_M]$, and the median-aggregated version is $\tilde{\Lambda}(\ell) = \text{med}_{w \in \mathcal{W}} \hat{\Lambda}_w(\ell)$.

A natural question arises: which learner’s results should the researcher report? When a single learner clearly dominates the $\tilde{\Lambda}$ ranking, one may report that learner’s inference with confidence. However, the apparent ranking may reflect finite-sample noise rather than genuine differences in population Λ values, particularly when many learners achieve similar performance. Chernozhukov et al. (2025a) acknowledge this issue and suggest two remedies for the case of two “near-winners”: reporting the union of their confidence intervals or applying a Bonferroni correction. However, neither the determination of which learners qualify as near-winners nor the extension to more than two near-winners is formalized. The next section proposes a data-driven solution to both problems via the model confidence set.

Formalizing the near-winner set requires comparing learner performance in a way that accounts for the multiplicity of comparisons and the dependence structure induced by evaluating all learners on the same data. Two natural approaches present themselves, but each faces obstacles in the GML setting. First, one could test pairwise hypotheses of the form $H_0 : \Lambda(\ell) = \Lambda(\ell')$ using a studentized test statistic. However, while Lemma D.1 of Chernozhukov et al. (2025a) establishes the asymptotic normality of the split-sample estimators $\hat{\theta}_{\ell w}$ conditional on the auxiliary sample, no analogous distributional result exists for $\hat{\Lambda}_w(\ell)$. The goodness-of-fit functional involves a squared coefficient and a variance term whose joint distribution—and dependence across learners—is not characterized by the existing theory. Second, one could apply a Romano-Wolf stepdown procedure (Romano and Wolf, 2007) directly to the parameter estimates across learners. This requires estimating the joint covariance of the median-aggregated test statistics, which in turn involves pooling influence functions across splits that are each valid only conditional on their respective auxiliary samples.

I instead take a different approach, treating the split-level performance values as a panel of loss observations and applying the model confidence set procedure of Hansen et al. (2011). The MCS avoids both difficulties: it operates on the split-level $\hat{\Lambda}_w(\ell)$ values directly, and requires only that the loss differentials across splits satisfy a law of large numbers and central limit theorem. As shown below, this requirement is satisfied by the structure of GML’s repeated sample splitting.

3 Model Confidence Set for GML

In this section I adapt the model confidence set of Hansen et al. (2011) to learner selection in Generic Machine Learning. A useful feature of the MCS framework is that it is intentionally agnostic about what constitutes a “model.” Forecasts, trading rules, treatment rules, and other competing procedures are all admissible objects of comparison. The object itself need not be associated with a structural population parameter. What the MCS requires is instead a stochastic loss process: for each object i , one observes losses L_{it} , forms relative losses $d_{ijt} = L_{it} - L_{jt}$, and assumes that the corresponding expected relative loss exists. The superior set is then defined in terms of expected loss, not in terms of whether any object is “true” in a structural sense.

This perspective is well suited to GML. In Hansen et al. the index t labels repeated realizations of the loss-generating environment. In the present setting, the analogous source of randomness is repeated sample splitting. The procedure described in Section 3.1 begins by setting aside a selection sample $\mathcal{D}^{\text{sel}}$ through a single outer split of the data; all learner comparisons in this section are made on that sample. Fix the realized selection sample $\mathcal{D}^{\text{sel}}$, and let A denote a random split drawn from the collection $\mathcal{A}$ of allowable auxiliary/main partitions of $\mathcal{D}^{\text{sel}}$ under the splitting scheme described below. For learner i , let $L_i(A; \mathcal{D}^{\text{sel}})$ denote its split-specific loss, and for learners i and j define the relative loss

$$d_{ij}(A; \mathcal{D}^{\text{sel}}) = L_i(A; \mathcal{D}^{\text{sel}}) - L_j(A; \mathcal{D}^{\text{sel}}).$$

The relevant analogue of Hansen et al.’s mean relative loss is therefore the conditional expectation

$$\mu_{ij}(\mathcal{D}^{\text{sel}}) = \mathbb{E}_A[d_{ij}(A; \mathcal{D}^{\text{sel}}) \mid \mathcal{D}^{\text{sel}}],$$

provided it is finite. This is the object used to compare learners throughout this section.

Accordingly, the MCS in the GML setting is interpreted conditionally on the realized selection sample. Define the conditional superior set by

$$\mathcal{M}^*(\mathcal{D}^{\text{sel}}) = \{i \in \mathcal{M}^0 : \mu_{ij}(\mathcal{D}^{\text{sel}}) \leq 0 \text{ for all } j \in \mathcal{M}^0\},$$

where $\mathcal{M}^0$ is the initial collection of candidate learners. Thus, a learner is “superior” if, under repeated admissible splits of the selection sample, it has no larger expected loss than any com-

petitor. This preserves the original MCS logic with a different loss-generating mechanism: there the randomness comes from the repeated realization indexed by t ; here it comes from the repeated random split A (which is indexed by w in the following sections). The remainder of this section defines the split-level loss used for GML learners and then applies the standard MCS logic to this conditional loss process.

3.1 Setup and Definitions

Fix the realized dataset $\mathcal{D}$, whose observations are i.i.d. draws, and the initial collection of candidate learners

$$\mathcal{L} = \{1, \dots, L\}.$$

The candidate collection is fixed and finite: the asymptotic statements below hold for fixed L and do not cover regimes in which the number of learners grows with the sample size. To make the coverage argument transparent, the procedure begins with a single outer split of the sample, generated independently of the data and fixed for the remainder of the analysis. The outer split partitions $\mathcal{D}$ into a selection sample $\mathcal{D}^{\text{sel}}$, used exclusively for learner selection, and an inference sample $\mathcal{D}^{\text{inf}}$, used exclusively for the reported inference. Let ρ^{sel} denote the fraction of observations allocated to selection, with $\rho^{\text{inf}} = 1 - \rho^{\text{sel}}$. Within each side, the procedure runs the standard GML splitting scheme of Section 2.1: the selection sample is repeatedly split into auxiliary and main subsamples in proportions ρ_A^{sel} and $1 - \rho_A^{\text{sel}}$, and the inference sample is likewise repeatedly split in proportions ρ_A^{inf} and $1 - \rho_A^{\text{inf}}$. The proportions need not be fixed, and there is a design trade-off between statistical power for learner selection (larger ρ^{sel}) and for the reported inference (larger ρ^{inf}). For a working example throughout the paper, I use $\rho^{\text{sel}} = 0.3$ and $\rho_A^{\text{sel}} = \rho_A^{\text{inf}} = 0.5$, preserving the equal auxiliary/main split of the original GML procedure on each side while allocating the smaller share of observations to selection.

Let $\mathcal{A}$ denote the collection of allowable auxiliary/main splits of the selection sample. Thus, each $A \in \mathcal{A}$ indexes a partition of $\mathcal{D}^{\text{sel}}$ into $(\mathcal{D}_A^{\text{sel}}(A), \mathcal{D}_M^{\text{sel}}(A))$ satisfying the splitting rule used by the procedure, including any design restrictions such as the fixed proportions above or stratification. Let A denote a generic random split taking values in $\mathcal{A}$. The MCS is applied to the loss process generated by this split randomness, conditional on the realized selection sample $\mathcal{D}^{\text{sel}}$.

For each realized split $A_w \in \mathcal{A}$, $w = 1, \dots, W^{\text{sel}}$, and learner $\ell \in \mathcal{L}$, the trained proxy $\hat{S}_{\ell,w}$ is obtained from $\mathcal{D}_{A,w}^{\text{sel}}$ following the GML procedure of Section 2.1, and the split-level goodness-of-fit value $\hat{\Lambda}_w(\ell)$ used by the MCS is computed by evaluating $\hat{S}_{\ell,w}$ on $\mathcal{D}_{M,w}^{\text{sel}}$. The selection stage is therefore entirely self-contained: no observation in $\mathcal{D}^{\text{inf}}$, and no object computed from $\mathcal{D}^{\text{inf}}$, enters the construction of the MCS. Conversely, no object computed on the selection sample—trained proxies included—enters the reported inference. The inference stage described in Section 4.3 applies the standard GML procedure to $\mathcal{D}^{\text{inf}}$ for each learner retained by the MCS, generating its own

collection of W^{inf} auxiliary/main splits, its own trained proxies, and its own WR estimates. The two split counts play different roles: the MCS coverage guarantee is asymptotic in W^{sel} , whereas the inference-stage guarantee holds for any fixed W^{inf} , as discussed in Section 4.3. In the working example, $W^{\text{sel}} = W^{\text{inf}} = 100$.

To apply the MCS framework, define the loss associated with learner ℓ in split w as the negative of the bias-corrected goodness-of-fit:

$$L_{\ell w} = -\hat{\Lambda}_w^{bc}(\ell), \quad \hat{\Lambda}_w^{bc}(\ell) = \left(\hat{\beta}_{2,\ell w}^2 - \hat{s}e_{\ell w}^2 \right) \hat{V}_{\ell w}, \quad (6)$$

where $\hat{\beta}_{2,\ell w}$ and $\hat{s}e_{\ell w}$ are computed by a WR regression on $\mathcal{D}_{M,w}^{\text{sel}}$ using the trained proxy $\hat{S}_{\ell,w}$, and $\hat{V}_{\ell w} = \hat{\text{Var}}(\hat{S}_{\ell,w}(Z) \mid \mathcal{D}_{M,w}^{\text{sel}})$ is the sample variance of the proxy predictions in the main portion of the selection sample for split w . The negation ensures that learners with better split-level goodness of fit have smaller loss, so that the MCS procedure retains the learners with the best conditional performance.

The motivation for the specific bias correction in (6)—why the raw squared coefficient $\hat{\beta}_{2,\ell w}^2$ can mechanically reward noisier learners—is treated as a loss-design question and deferred to Section 4.2. The MCS validity in Section 4.1 holds for any loss satisfying the regularity conditions stated therein.²

For learners $i, j \in \mathcal{L}$, define the generic relative loss

$$d_{ij}(A; \mathcal{D}^{\text{sel}}) = L_i(A; \mathcal{D}^{\text{sel}}) - L_j(A; \mathcal{D}^{\text{sel}}),$$

and its realized analogue in split w ,

$$d_{ij,w} = L_{i,w} - L_{j,w} = \hat{\Lambda}_w^{bc}(j) - \hat{\Lambda}_w^{bc}(i). \quad (7)$$

The corresponding conditional mean relative loss is

$$\mu_{ij}(\mathcal{D}^{\text{sel}}) = \mathbb{E}_A[d_{ij}(A; \mathcal{D}^{\text{sel}}) \mid \mathcal{D}^{\text{sel}}].$$

Learner i is preferred to learner j when $\mu_{ij}(\mathcal{D}^{\text{sel}}) < 0$, meaning that learner i has lower expected loss under repeated admissible splits of the selection sample. Accordingly, define the conditional superior set by

$$\mathcal{M}^*(\mathcal{D}^{\text{sel}}) = \{i \in \mathcal{L} : \mu_{ij}(\mathcal{D}^{\text{sel}}) \leq 0 \text{ for all } j \in \mathcal{L}\}. \quad (8)$$

When a unique best learner exists, $\mathcal{M}^*(\mathcal{D}^{\text{sel}})$ is a singleton; when several learners have the same conditional expected loss, $|\mathcal{M}^*(\mathcal{D}^{\text{sel}})| > 1$.

²A researcher who prefers $-\hat{\Lambda}_\ell$ as the user-specified loss is free to do so, and the procedure remains valid for the superior set induced by that choice. Note that different loss criterion may induce different superior sets.

For any candidate set $\mathcal{M} \subseteq \mathcal{L}$, the null hypothesis of equal conditional performance is

$$H_{0,\mathcal{M}}(\mathcal{D}^{\text{sel}}) : \mu_{ij}(\mathcal{D}^{\text{sel}}) = 0 \quad \text{for all } i, j \in \mathcal{M}. \quad (9)$$

3.2 Equivalence Test

For a candidate set $\mathcal{M}$ with $|\mathcal{M}| = m$, define the relative sample loss of learner i against the average loss in $\mathcal{M}$ by

$$\bar{d}_{i\cdot} = \frac{1}{W^{\text{sel}}} \sum_{w=1}^{W^{\text{sel}}} d_{i\cdot,w}, \quad d_{i\cdot,w} = L_{i,w} - \frac{1}{m} \sum_{j \in \mathcal{M}} L_{j,w}. \quad (10)$$

Thus, $\bar{d}_{i\cdot}$ is the empirical counterpart, across the W^{sel} realized splits, to the conditional average excess loss of learner i relative to the mean loss in $\mathcal{M}$. Its estimated variance is

$$\hat{\text{Var}}(\bar{d}_{i\cdot}) = (W^{\text{sel}})^{-1} (W^{\text{sel}} - 1)^{-1} \sum_{w=1}^{W^{\text{sel}}} (d_{i\cdot,w} - \bar{d}_{i\cdot})^2.$$

The associated t -statistic for learner i is

$$t_{i\cdot} = \frac{\bar{d}_{i\cdot}}{\sqrt{\hat{\text{Var}}(\bar{d}_{i\cdot})}}, \quad (11)$$

and the equivalence test statistic is

$$T_{\max,\mathcal{M}} = \max_{i \in \mathcal{M}} t_{i\cdot}. \quad (12)$$

A large value of $T_{\max,\mathcal{M}}$ indicates that at least one learner in $\mathcal{M}$ has substantially larger average loss than the others, providing evidence against $H_{0,\mathcal{M}}(\mathcal{D}^{\text{sel}})$.

The equivalence test rejects $H_{0,\mathcal{M}}(\mathcal{D}^{\text{sel}})$ at level α_1 when $T_{\max,\mathcal{M}}$ exceeds the $(1 - \alpha_1)$ -quantile of its bootstrap distribution, computed as described in Section 3.3. Upon rejection, the learner contributing most to the test statistic is eliminated according to the rule

$$e_{\mathcal{M}} = \arg \max_{i \in \mathcal{M}} t_{i\cdot}. \quad (13)$$

3.3 Bootstrap Implementation

To justify the ordinary nonparametric bootstrap, I impose an explicit sampling assumption on the repeated splits used to construct the empirical loss sequence.

Assumption 1. (*Repeated split sampling*) Conditional on the realized selection sample $\mathcal{D}^{\text{sel}}$, the

realized splits

$$A_1, \dots, A_{W^{\text{sel}}} \stackrel{\text{i.i.d.}}{\sim} U(\mathcal{A}),$$

where $U(\mathcal{A})$ denotes the discrete uniform distribution on the collection of allowable splits.

Assumption 1 is mild in the present setting. It states that each split is generated independently from the same splitting rule. This is the natural probabilistic description of repeated sample splitting in practice, and it is only invoked here to justify the use of the standard nonparametric bootstrap. Conditional on the selection sample, Assumption 1 dictates that the split-level loss observations $\{L_{\ell w}\}_{w=1}^{W^{\text{sel}}}$ and the loss differentials $\{d_{ij,w}\}_{w=1}^{W^{\text{sel}}}$ are i.i.d.

Critical values for $T_{\max, \mathcal{M}}$ are obtained via a null-centered nonparametric bootstrap. For the current candidate set $\mathcal{M}$, first center each learner's observed relative-to-average loss series:

$$\tilde{d}_{i\cdot, w} = d_{i\cdot, w} - \bar{d}_{i\cdot}, \quad i \in \mathcal{M}, \quad w = 1, \dots, W^{\text{sel}},$$

where

$$\bar{d}_{i\cdot} = \frac{1}{W^{\text{sel}}} \sum_{w=1}^{W^{\text{sel}}} d_{i\cdot, w}.$$

By construction,

$$\frac{1}{W^{\text{sel}}} \sum_{w=1}^{W^{\text{sel}}} \tilde{d}_{i\cdot, w} = 0$$

for every $i \in \mathcal{M}$.

For each bootstrap draw $b = 1, \dots, B$:

1. Draw W^{sel} split indices

$$I_1^{(b)}, \dots, I_{W^{\text{sel}}}^{(b)}$$

independently with replacement from $\{1, \dots, W^{\text{sel}}\}$.

2. Construct the bootstrap relative-loss series

$$d_{i\cdot, w}^{*(b)} = \tilde{d}_{i\cdot, I_w^{(b)}}, \quad i \in \mathcal{M}, \quad w = 1, \dots, W^{\text{sel}}.$$

The same resampled split index is applied to all learners, thereby retaining the joint empirical distribution of their split-level relative losses.

3. Compute the bootstrap means

$$\bar{d}_{i\cdot}^{*(b)} = \frac{1}{W^{\text{sel}}} \sum_{w=1}^{W^{\text{sel}}} d_{i\cdot, w}^{*(b)},$$

and their estimated variances

$$\hat{\text{Var}}^{*(b)}\left(\bar{d}_{i\cdot}^{*(b)}\right) = \frac{1}{W^{\text{sel}}(W^{\text{sel}} - 1)} \sum_{w=1}^{W^{\text{sel}}} \left(d_{i\cdot,w}^{*(b)} - \bar{d}_{i\cdot}^{*(b)}\right)^2.$$

The corresponding bootstrap t -statistics and equivalence-test statistic are

$$t_{i\cdot}^{*(b)} = \frac{\bar{d}_{i\cdot}^{*(b)}}{\sqrt{\hat{\text{Var}}^{*(b)}\left(\bar{d}_{i\cdot}^{*(b)}\right)}}, \quad T_{\max, \mathcal{M}}^{*(b)} = \max_{i \in \mathcal{M}} t_{i\cdot}^{*(b)}.$$

The bootstrap p -value for the equivalence test on $\mathcal{M}$ is

$$\hat{p}_{H_0, \mathcal{M}} = \frac{1}{B} \sum_{b=1}^B \mathbb{1}\left(T_{\max, \mathcal{M}}^{*(b)} \geq T_{\max, \mathcal{M}}\right). \quad (14)$$

Centering the observed relative-loss series before resampling imposes the null hypothesis within the bootstrap procedure. Under $H_{0, \mathcal{M}}(\mathcal{D}^{\text{sel}})$, each learner's conditional mean excess loss relative to the average learner in $\mathcal{M}$ is zero. If one were instead to bootstrap the uncentered relative-loss series, the bootstrap statistic would be centered around the empirical alternative whenever the observed sample favors one learner over another, yielding a distorted reference distribution for the test. Subtracting the observed sample means removes this empirical advantage and makes the bootstrap sample mimic the null of equal conditional performance.

Importantly, the observed relative-loss series is centered once, before the bootstrap draws are generated. Each bootstrap resample is not recentered around its own bootstrap mean, since doing so would force every bootstrap numerator to equal zero identically. Equivalently, one could resample the original relative-loss vectors and use $\bar{d}_{i\cdot}^{*(b)} - \bar{d}_{i\cdot}$ in the numerator of the bootstrap t -statistic. This follows the unrestricted bootstrap logic in Hansen et al.: the goal is to approximate the null distribution of the equivalence test rather than to preserve the empirical ranking in the bootstrap resamples.³

3.4 The MCS Algorithm

The model confidence set $\hat{\mathcal{M}}_{1-\alpha_1}$ is constructed by the following sequential procedure at significance level α_1 :

Step 0. Set $\mathcal{M} = \mathcal{L}$.

³This null-centering step is conceptually distinct from the bias correction in (6). The bias correction is built into the loss itself so that noisier learners are not mechanically favored, whereas null-centering is a separate bootstrap device used only to obtain valid critical values under $H_{0, \mathcal{M}}(\mathcal{D}^{\text{sel}})$.

Step 1. Test $H_{0,\mathcal{M}}(\mathcal{D}^{\text{sel}})$ using $T_{\max,\mathcal{M}}$ at level α_1 .

Step 2. If $H_{0,\mathcal{M}}(\mathcal{D}^{\text{sel}})$ is not rejected, set $\hat{\mathcal{M}}_{1-\alpha_1} = \mathcal{M}$ and stop. If $H_{0,\mathcal{M}}(\mathcal{D}^{\text{sel}})$ is rejected, eliminate the learner $e_{\mathcal{M}} = \arg \max_{i \in \mathcal{M}} t_i$ and return to Step 1 with the reduced set.

The elimination rule $e_{\mathcal{M}} = \arg \max_{i \in \mathcal{M}} t_i$ removes the learner contributing most to the rejection, and satisfies the coherency condition of Hansen et al. (2011) (Proposition 1) with the $T_{\max,\mathcal{M}}$ test.

Each learner ℓ receives an MCS p -value $\hat{p}_\ell = \max_{k \leq k(\ell)} \hat{p}_{H_{0,\mathcal{M}_k}}$, where $k(\ell)$ is the elimination step at which ℓ is removed (or the final step if ℓ survives). By convention, the last surviving learner receives $\hat{p}_\ell = 1$. Learner ℓ belongs to $\hat{\mathcal{M}}_{1-\alpha_1}$ if and only if $\hat{p}_\ell > \alpha_1$.

4 Regularity Conditions and Loss Design

The MCS procedure of Section 3 requires conditions on the loss differentials to deliver its coverage guarantee. This section states those conditions (Section 4.1), argues that the bias-corrected loss in (6) is a well-designed criterion for learner comparison (Section 4.2), and derives the overall coverage of the two-stage procedure (Section 4.3). Throughout this section, $\Pr(\cdot)$ denotes joint probability over the data-generating process and all split randomization, including the outer split; $\Pr(\cdot \mid \mathcal{D}^{\text{sel}})$, $\mathbb{E}[\cdot \mid \mathcal{D}^{\text{sel}}]$, and $\text{Var}(\cdot \mid \mathcal{D}^{\text{sel}})$ refer to the selection-stage split randomness alone, conditional on the realized selection sample.

4.1 Regularity of Loss Differentials

The i.i.d. split sampling in Assumption 1 provides the core probabilistic structure needed for the MCS. Two additional regularity conditions, which are standard for least-squares estimators computed on a fixed dataset, complete the requirements.

Assumption 2. (*Regularity of split-level losses.*) Conditional on $\mathcal{D}^{\text{sel}}$:

- (i) For each learner $\ell \in \mathcal{L}$ and each admissible split $A \in \mathcal{A}$, the split-level loss $L_\ell(A; \mathcal{D}^{\text{sel}})$ is finite.
- (ii) For every $\mathcal{M} \subseteq \mathcal{L}$ with $|\mathcal{M}| \geq 2$ and every $i \in \mathcal{M}$, the relative-to-average loss in (10) is nondegenerate: $0 < \text{Var}(d_{i,w} \mid \mathcal{D}^{\text{sel}}) < \infty$.

Because the collection $\mathcal{A}$ of admissible splits is finite conditional on $\mathcal{D}^{\text{sel}}$, the split distribution has finite support, so Assumption 2(i) implies that the losses and loss differentials possess finite moments of all orders. Condition (i) itself holds whenever the WR regression is well defined in every admissible split, which requires only the absence of perfect collinearity in the design. Assumption 2(ii) is imposed directly on the denominator of the statistic t_i and rules out the degenerate case in which a learner's loss coincides with the within-set average loss in every split. Pairwise nondegeneracy,

$\text{Var}(d_{ij,w} \mid \mathcal{D}^{\text{sel}}) > 0$, is not sufficient for this: with three learners whose centered losses satisfy $L_{1,w} = 0$, $L_{2,w} = X_w$, and $L_{3,w} = -X_w$ for nondegenerate X_w , every pairwise differential has positive variance while $d_{1,w} = 0$ identically. Condition (ii) is quantified over all candidate sets with at least two learners because the sequential algorithm operates on shrinking candidate sets; singletons are excluded since $d_{i,w} = 0$ identically when $\mathcal{M} = \{i\}$, and the algorithm terminates before testing a singleton. The condition holds in any practical implementation in which distinct learning algorithms produce non-identical predictions.

Together, Assumptions 1 and 2 are stronger than the corresponding assumption of Hansen et al. (2011)⁴ which only places mixing order and stationary conditions on the relative loss variables. The i.i.d. structure of repeated sample splitting makes the GML context a special case where mixing and stationarity requirements are naturally satisfied rather than imposed as a restriction.

Proposition 1. *Let Assumptions 1 and 2 hold. Then the MCS procedure of Section 3, applied with the equivalence test $T_{\max, \mathcal{M}}$ and elimination rule $e_{\mathcal{M}} = \arg \max_{i \in \mathcal{M}} t_i$, satisfies*

- (i) $\liminf_{W^{\text{sel}} \rightarrow \infty} \Pr\left(\mathcal{M}^*(\mathcal{D}^{\text{sel}}) \subseteq \hat{\mathcal{M}}_{1-\alpha_1} \mid \mathcal{D}^{\text{sel}}\right) \geq 1 - \alpha_1$, and
- (ii) $\lim_{W^{\text{sel}} \rightarrow \infty} \Pr\left(i \in \hat{\mathcal{M}}_{1-\alpha_1} \mid \mathcal{D}^{\text{sel}}\right) = 0 \quad \forall i \notin \mathcal{M}^*(\mathcal{D}^{\text{sel}})$.

Proof. See Appendix B.

4.2 Loss Design

Proposition 1 guarantees that the MCS contains the conditionally superior learners for *any* loss satisfying Assumptions 1 and 2. This guarantee is unconditional on the choice of criterion. If a researcher uses a poorly designed loss, the MCS is still a valid confidence set—it simply covers the “best” learners under a potentially meaningless ranking. The practical question is whether the loss reflects an intended notion of learner quality.

The raw goodness-of-fit $\hat{\Lambda}_w(\ell) = \hat{\beta}_{2,\ell w}^2 \hat{V}_{\ell w}$ is a natural candidate but has a deficiency. $\mathbb{E}[\hat{\beta}_{2,\ell w}^2 \mid \mathcal{D}^{\text{sel}}]$ overstates the squared signal because it includes the across-split variance of the coefficient. By the identity $\mathbb{E}[X^2] = (\mathbb{E}[X])^2 + \text{Var}(X)$,

$$\mathbb{E}[\hat{\beta}_{2,\ell w}^2 \mid \mathcal{D}^{\text{sel}}] = \left(\mathbb{E}[\hat{\beta}_{2,\ell w} \mid \mathcal{D}^{\text{sel}}]\right)^2 + \text{Var}(\hat{\beta}_{2,\ell w} \mid \mathcal{D}^{\text{sel}}),$$

where all expectations are over the split randomness conditional on $\mathcal{D}^{\text{sel}}$. The variance term is learner-specific: a learner whose coefficient fluctuates more across splits receives a mechanical boost to the raw criterion, even if its average signal is no stronger. Two learners with identical average signals but different across-split variances would be ranked differently.

⁴The corresponding assumption is their Assumption 2 (p 463).

The bias-corrected loss in (6) addresses this by subtracting the squared within-split standard error, $\hat{\text{se}}_{\ell w}^2$. This standard error is obtained from the WR regression on the main portion of the selection sample, $\mathcal{D}_{M,w}^{\text{sel}}$, and consistently estimates the sampling variance of $\hat{\beta}_{2,\ell w}$ conditional on the auxiliary portion and the trained proxy, holding both fixed. By the law of total variance, $\text{Var}(\hat{\beta}_{2,\ell w} \mid \mathcal{D}^{\text{sel}})$ decomposes into this within-split component and a residual reflecting variation across auxiliary samples in the BLP coefficient associated with the realized proxy. The correction $\hat{\beta}_{2,\ell w}^2 - \hat{\text{se}}_{\ell w}^2$ removes the within-split component exactly in expectation but leaves the auxiliary-sample residual. Whether this residual differs across learners by enough to reverse the ranking induced by an idealized conditional goodness-of-fit measure is a loss-design question taken up in Appendix C; it does not affect the validity of the MCS.

Two additional points are worth noting. First, the GATES-based goodness-of-fit $\bar{\Lambda}_w(\ell) = G^{-1} \sum_{k=1}^G \hat{\gamma}_{k,\ell w}^2$ admits the same bias-correction logic: each $\hat{\gamma}_{k,\ell w}^2$ is inflated by $\text{Var}(\hat{\gamma}_{k,\ell w} \mid \mathcal{D}^{\text{sel}})$, and the correction $\hat{\gamma}_{k,\ell w}^2 - \hat{\text{se}}_{k,\ell w}^2$ removes it. Since $\bar{\Lambda}_w^{bc}(\ell)$ is a sum of corrected squared coefficients rather than a product with a separate variance term, the cross-term analyzed in Appendix C does not arise, making the GATES-based criterion somewhat cleaner from a loss-design perspective. The auxiliary-sample residual identified above is, however, present in both criteria.

Second, and importantly, even if the cross-term in the BLP-based criterion does not fully vanish at a given sample size, the MCS coverage guarantee from Proposition 1 is unaffected. The coverage holds for any loss satisfying the regularity conditions; the loss-design analysis concerns only the interpretability of the resulting $\mathcal{M}^*(\mathcal{D}^{\text{sel}})$, not the validity of the procedure. The simulation results in Section 5 document size control under the homogeneous DGP and meaningful power under the heterogeneous DGP for both BLP and GATES criteria, indicating that the bias-corrected losses behave as expected in the configurations studied.

4.3 Two-Stage Inference

With the model confidence set in hand, inference on the target parameter θ_ℓ (e.g., $\beta_{2,\ell}$ or a GATES contrast $\gamma_G - \gamma_1$) proceeds over $\hat{\mathcal{M}}_{1-\alpha_1}$ rather than a single selected learner or the full candidate set. This subsection derives the overall coverage of the two-stage procedure—MCS selection followed by median-aggregated inference—and clarifies how the asymptotic and finite-sample components interact. In what follows, I take the GML framework of Chernozhukov et al. (2025a) (CDDF) as given and inherit only the minimal regularity conditions such that the coverage results remain applicable to a wide range of use cases.⁵

The outer split of Section 3.1 plays a central role here. The MCS is constructed entirely from $\mathcal{D}^{\text{sel}}$, while the median-aggregated confidence intervals below are constructed entirely from $\mathcal{D}^{\text{inf}}$, using its own independent collection of auxiliary/main splits. Under i.i.d. sampling, the two samples

⁵Minimal regularity conditions are R1 and R3 of CDDF.

are independent, and the split randomizations on each side are generated independently of the data and of one another. Every selection-stage object—the losses, the model confidence set, and any learner chosen from it—is therefore independent of every inference-stage object. This independence is what allows the pointwise CDDF coverage guarantee to transfer to a learner determined by the selection stage. Holding out a dedicated sample for selection is the classical device for honest inference after selection (see, e.g., the discussion of data splitting in Fithian et al., 2017), and the posture is shared by the single-split alternatives to repeated-split GML (Wager, 2025; Imai and Li, 2025): the selection step is carried out on observations that never enter the reported inference.

4.3.1 Median-aggregated confidence intervals

The inference stage generates its own finite collection $\mathcal{A}^{\text{inf}}$ of W^{inf} auxiliary/main splits of $\mathcal{D}^{\text{inf}}$, drawn independently of the data. Following Definition 4.1 of CDDF, the aggregation quantiles and the inferential target are defined over this realized collection: throughout, θ_ℓ denotes the median estimand $\text{Med}[\theta_{\ell,A} \mid \mathcal{D}^{\text{inf}}, \mathcal{A}^{\text{inf}}]$ with $A \sim U(\mathcal{A}^{\text{inf}})$, which is the target of Theorem 4.3(i) of CDDF. For each learner $\ell \in \hat{\mathcal{M}}_{1-\alpha_1}$, within-split confidence intervals are constructed at nominal level $\zeta = \alpha_2/2$ from the WR regression on $\mathcal{D}_{M,w}^{\text{inf}}$ and median-aggregated across the W^{inf} inference-stage splits:

$$\tilde{I}_{\theta_\ell}(\alpha_2) = \left[\text{med}_{1 \leq w \leq W^{\text{inf}}} \left(\hat{\theta}_{\ell w} - z_{1-\zeta/2} \hat{\text{se}}_{\ell w} \right), \text{med}_{1 \leq w \leq W^{\text{inf}}} \left(\hat{\theta}_{\ell w} + z_{1-\zeta/2} \hat{\text{se}}_{\ell w} \right) \right]. \quad (15)$$

By Theorems 4.2-4.3 of CDDF, each within-split interval at level ζ produces a median-aggregated interval with coverage at least $1 - 2\zeta = 1 - \alpha_2$, provided the within-split intervals are conditionally valid.⁶ The within-split validity follows from the approximate normality of the WR regression in (3) on $(1 - \rho_A^{\text{inf}})\rho^{\text{inf}}N$ observations (Lemma D.1 of CDDF), which is an asymptotic-in- n result. Thus, the CDDF coverage guarantee

$$\Pr(\theta_\ell \in \tilde{I}_{\theta_\ell}(\alpha_2)) \geq 1 - \alpha_2 - o(1) \quad (16)$$

holds under minimal regularity conditions for any fixed W^{inf} : because both the target and the aggregation quantiles are defined over the realized collection $\mathcal{A}^{\text{inf}}$, the quantile-aggregation argument of CDDF applies at the number of splits actually used, and no limit in W^{inf} is required. The bound does implicitly require $(1 - \rho_A^{\text{inf}})\rho^{\text{inf}}N$ to be large enough for the within-split CLT to provide adequate approximation. CDDF note that the coverage of (16) improves to $1 - \zeta - o(1)$ under an additional concentration condition (CDDF R2), and their numerical experiments result in coverage exceeding $1 - \zeta$. The simulation results in Section 5 also lend mild support to this

⁶Levels are presented here with α_2 as the target inferential significance level. For example, if one wants 90% final confidence intervals for the median-aggregated point estimate, then one must construct 95% CI bounds in each split; aggregating then yields the desired 90% interval. This example corresponds to setting $\alpha_2 = 0.10$ and $\zeta = 0.05$.

view, however, the evidence for $1 - \zeta$ coverage of the median-aggregated CI remains empirical. In practice, the prevailing assumption is that only CDDF’s minimal regularity conditions hold. The popular **GenericML** package (Welz et al., 2022) and the study used for the application in Section 6 (Bryan et al., 2024) both make this assumption.⁷ Accordingly, all coverage statements in this paper use the conservative bound of (16), which requires only CDDF’s minimal regularity conditions and matches the convention adopted in applied practice.

4.3.2 Union inference over the MCS

The union confidence interval over the model confidence set is

$$\tilde{I}^U(\alpha_2) = \left[\min_{\ell \in \hat{\mathcal{M}}_{1-\alpha_1}} \tilde{l}_{\theta_\ell}, \max_{\ell \in \hat{\mathcal{M}}_{1-\alpha_1}} \tilde{u}_{\theta_\ell} \right], \quad (17)$$

where $\tilde{l}_{\theta_\ell}$ and $\tilde{u}_{\theta_\ell}$ denote the lower and upper bounds of $\tilde{I}_{\theta_\ell}(\alpha_2)$. If $|\hat{\mathcal{M}}_{1-\alpha_1}| = 1$, the union interval reduces to the single learner’s median-aggregated interval with no penalty. If $|\hat{\mathcal{M}}_{1-\alpha_1}| > 1$, the union interval widens to cover all surviving learners’ intervals.⁸

4.3.3 Combined coverage

The two-stage procedure draws on two sources of coverage: the MCS guarantee from Proposition 1 and the median-aggregated CI guarantee from CDDF. Combining these yields the main inferential result of the paper. The outer split is essential to the argument: because the model confidence set is a function of the selection stage alone and the median-aggregated intervals are functions of the inference stage alone, the pointwise CDDF coverage in (16) transfers to the conditionally best learner even though that learner’s identity is unknown to the researcher and its membership in the reported set is determined by the data.

Theorem 1. *Let the conditions of Proposition 1 and CDDF Theorem 4.3(i) hold, and suppose the outer split of Section 3.1 is applied. Let ℓ^* denote any element of $\mathcal{M}^*(\mathcal{D}^{\text{sel}})$, fixed by an arbitrary deterministic tie-breaking rule, and let $n = (1 - \rho_A^{\text{inf}})\rho^{\text{inf}}N$ denote the inference-stage main-sample size. Then, taking limits sequentially,*

$$\liminf_{n \rightarrow \infty} \liminf_{W^{\text{sel}} \rightarrow \infty} \Pr(\theta_{\ell^*} \in \tilde{I}^U(\alpha_2)) \geq 1 - \alpha_1 - \alpha_2.$$

⁷As of version 0.2.3, the strategy of **GenericML** is to median aggregate upper and lower bounds of a $1 - \alpha$ confidence interval, but report the confidence level as $1 - 2\alpha$ in the output (see construction of the `accessor_output()` function on the GitHub repo). Bryan et al. (2024, p. 2842) states that, among other adjustments, that for a $1 - \alpha$ CI, they report the median of $1 - 2\alpha$ CI boundaries. This is a small mistake in the text since median aggregation should *cost* coverage. Their code does not commit this error. Had either of these examples relied on CDDF’s R2 or empirical results, the choice would have been to aggregate $1 - \alpha$ confidence intervals and report as is.

⁸If simultaneous coverage of all θ_ℓ , $\ell \in \hat{\mathcal{M}}_{1-\alpha_1}$, is desired rather than coverage of the conditionally best learner’s parameter, one may replace $\tilde{I}_{\theta_\ell}(\alpha_2)$ with Bonferroni-corrected intervals at level $\alpha_2/|\hat{\mathcal{M}}_{1-\alpha_1}|$.

Proof. See Appendix B. □

Remark 1. *Theorem 1 covers the parameter of the conditionally best learner, whose identity is unknown. If interest instead centers on the parameter of the specific learner the researcher reports—say, the lowest-loss member of $\hat{\mathcal{M}}_{1-\alpha_1}$ —then the independence of the two stages alone delivers $\Pr(\theta_{\hat{\ell}} \in \tilde{I}_{\theta_{\hat{\ell}}}(\alpha_2)) \geq 1 - \alpha_2 - o(1)$ for any selection rule $\hat{\ell}$ that depends only on the selection stage, with no α_1 cost. The α_1 budget is therefore the price of the stronger claim: with probability at least $1 - \alpha_1$, the reported set contains the conditionally superior learners, so the union interval covers the best learner’s parameter rather than merely the parameter of whichever learner happened to be selected.*

The bound inherits asymptotic qualifications from both of its components, and the sequential limits in the theorem statement track them separately. The MCS coverage from Proposition 1 is asymptotic in W^{sel} : it requires the number of selection-stage splits to be large enough for the bootstrap to approximate the null distribution of $T_{\max, \mathcal{M}}$. The CDDF coverage is asymptotic in the inference-stage main-sample size $n = (1 - \rho_A^{\text{inf}})\rho^{\text{inf}}N$: it requires the main sample to be large enough for the within-split CLT to apply. No limit is taken in W^{inf} , since the inference-stage guarantee (16) holds at the realized number of splits. The n -asymptotic component is shared with standard GML inference—any application of CDDF relies on the same within-split CLT—so the only new asymptotic layer introduced by the MCS is $W^{\text{sel}} \rightarrow \infty$, which is a mild requirement given the typical use of $W^{\text{sel}} = 100$ or more splits. The finite-sample performance of the combined procedure is assessed in Section 5.

The outer split introduces a source of randomness that the repeated inner splits do not average over: a different outer split allocates different observations to selection versus inference. The coverage guarantee of Theorem 1 already accounts for this randomness, because the outer split is generated independently of the data and the probability statement is unconditional over both the data and all split randomization. In practice, the outer split should be generated once and fixed ex ante, for example by committing to a randomization seed before the analysis, which forecloses any search over outer splits. A researcher who wishes to demonstrate that the reported conclusions are not an artifact of the particular outer split can repeat the full procedure across several outer splits and report the stability of the selected set and of the resulting intervals as a descriptive robustness exercise, in the same spirit as the repeated-split robustness that motivates GML itself. No inferential claim is attached to these replications; the formal guarantee belongs to the pre-committed split.⁹

It is worth comparing Theorem 1 to the alternatives available without the MCS. If a researcher applies a Bonferroni correction over all L candidate learners, each learner’s CI is inflated by L and

⁹A formal aggregation across outer splits is also available: repeating the outer split several times and median-aggregating the resulting union intervals remains valid under arbitrary dependence across outer splits by the quantile-aggregation results of CDDF, at the cost of doubling the total error budget to $2(\alpha_1 + \alpha_2)$. A systematic comparison of this variant with the fixed-outer-split procedure is left for future work.

the resulting coverage is $1 - \alpha_2 - o(1)$ with no selection error. If the researcher instead takes the union of all L learners' CIs, the coverage is again $1 - \alpha_2 - o(1)$, but the interval spans the most extreme bounds across all learners, including those with poor proxies. Both approaches avoid the α_1 cost of selection but pay for it with wider intervals. The MCS introduces a selection error of α_1 but restricts the union or Bonferroni correction to $|\hat{\mathcal{M}}_{1-\alpha_1}|$ learners rather than L . When the MCS selects a singleton, there is no width penalty at all: the interval is identical to that of naive top-1 selection, but with valid $1 - \alpha_1 - \alpha_2$ coverage. The researcher controls the tradeoff between selection risk and interval width through the choice of α_1 .

These comparisons also clarify how the MCS improves upon current reporting practice. A researcher who finds the full- L correction too conservative may be tempted to restrict it to a subset $\mathcal{L}' \subset \mathcal{L}$ of “near-winners,” since $|\mathcal{L}'| < L$ would tighten intervals. But this reintroduces the problem: the choice of which learners belong in $\mathcal{L}'$ is subjective, and no theoretical guarantee covers the selection step. The result is a tension between wanting a large candidate set (to avoid missing the best learner) and wanting a small correction factor (to preserve power).

The MCS resolves this tension. The researcher may begin with any fixed finite collection of candidate learners, however large—a prudent strategy when there is no *a priori* reason to expect one algorithm to dominate on a given dataset. The MCS then provides a data-driven reduction to $\hat{\mathcal{M}}_{1-\alpha_1}$, eliminating underperforming learners with a formal testing procedure rather than subjective judgment. Inference over the surviving set carries the coverage guarantee of Theorem 1, regardless of how large the initial candidate set was.

5 Simulation Results

The data generating process follows Wager (2025) and the rejoinder of Chernozhukov et al. (2025b). Covariates $Z_i \in \mathbb{R}^{10}$ are drawn i.i.d. from $\text{Uniform}(-1, 1)^{10}$, treatment is assigned as $D_i \sim \text{Bernoulli}(0.5)$, and outcomes are generated as

$$Y_i = D_i \cdot \tau(Z_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 1), \quad (18)$$

with $\tau(z) = \max(0, z_1)$ under the heterogeneous design and $\tau(z) = 0$ under the homogeneous design. The sample size is $n = 1000$. This DGP provides a useful benchmark: the CATE is a simple, piecewise-linear function of a single covariate, producing moderate treatment effect heterogeneity that is detectable but not overwhelming. Only one of the ten covariates is relevant, creating a setting in which learner performance is expected to vary meaningfully across ML methods.

I run 10,000 Monte Carlo replications for both homogeneous and heterogeneous versions of (18) and record rejection rates from a test $H_0 : \beta_2 = 0$, where β_2 is the parameter in (3). Recall that $s_0(Z)$ is constant under the homogeneous DGP, and thus $\text{Cov}[s_0(Z), S(Z)] \approx 0$ which implies

failure to reject $\beta_2 = 0$. Under the heterogeneous DGP, the theoretical value for β_2 is unknown. However, because heterogeneity is present, a good proxy will uncover the systematic variation in $s_0(Z)$ and result in a nonzero estimate of β_2 .¹⁰ Unless stated otherwise, the outer split allocates $\rho^{\text{sel}} = 0.3$ of the observations to the selection sample, and both stages use equal auxiliary/main splits, $\rho_A^{\text{sel}} = \rho_A^{\text{inf}} = 0.5$.

A total of seven learners were used: partial least squares, random forest, lasso, trees, generalized linear model, k-nearest neighbors, and causal forest. All are implemented in R via the `caret` package (Kuhn and Max, 2008) except causal forest, for which I use the `grf` package (Tibshirani et al., 2024; Athey et al., 2019).¹¹ For all but causal forest, the CATE predictions are generated via the “T-learner” approach (see Künzel et al. (2019) for more details on meta learners). This involves simply fitting ℓ to $\mathcal{D}_A \mid D_i = 0$ to obtain $\mu_{(0)\ell}(Z)$ and to $\mathcal{D}_A \mid D_i = 1$ to obtain $\mu_{(1)\ell}(Z)$.¹² CATE predictions are obtained by generating predictions of both $\mu_{(0)\ell}(Z)$ and $\mu_{(1)\ell}(Z)$ in the $\mathcal{D}_M$ sample and taking the difference, $S_\ell(Z) = \hat{\mu}_{(1)\ell}(Z) - \hat{\mu}_{(0)\ell}(Z)$.¹³ Causal forests do not require the T-learner as the method is purpose-built for learning the CATE. As mentioned in the introduction, it is rarely the case that a researcher has an *a priori* about which learning algorithms are best suited to uncovering patterns in the true CATE.¹⁴ Beginning with a relatively large set of candidate learners casts a wide net and offers a prudent attempt at uncovering these patterns.

To illustrate the challenge of learner selection, Table 1 reports the mean rank and standard deviation of each learner’s bias-corrected $\tilde{\Lambda}^{bc}$ across Monte Carlo replications. Under the homogeneous DGP, all mean ranks cluster near the midpoint of 4 with standard deviations around 2, indicating that the ranking is effectively random. A researcher selecting the “best” learner in this setting is choosing based on noise. Under the heterogeneous DGP, a clear hierarchy emerges: lasso ranks first on average (mean rank 1.41), with causal forest second (2.46) and the remaining learners more variable in placement, but with sufficient variability that the identity of the second- or third-best learner shifts across replications.

Figure 1 displays the distribution of $\tilde{\Lambda}^{bc}$ by learner. Under the homogeneous design, all learners produce indistinguishable performance values. Under the heterogeneous design, lasso and causal forest separate from the remaining learners, but the overlap in distributions underscores the difficulty of drawing a sharp boundary between near-winners and clearly inferior candidates.

Table 3 displays the simulation results. Size and power correspond to rejection rates under the homogeneous and heterogeneous processes, respectively. The first row contains the results from

¹⁰The sign of β_2 is positive given a good proxy since $\text{Cov}[s_0(Z), S(Z)] < 0$ would be a case where $S(Z)$ predicts heterogeneity in the wrong direction.

¹¹For `caret` learners: the regression versions of `pls`, `rf`, `lasso`, `rpart`, `glm`, `knn`. `grf::causal_forest()` is used to implement the causal forest of Athey et al. (2019).

¹²Note that $\mu_{(0)\ell}(Z)$ is the BCA, $B_\ell(Z)$.

¹³The T-learner is a common method employed in practice and the one utilized in the popular R package, `GenericML` (Welz et al., 2022).

¹⁴One notable exception is Deryugina et al. (2019).

Table (1) Mean rank and standard deviation of $\tilde{\Lambda}^{bc}$ rank by learner

	Homogeneous		Heterogeneous	
	Mean	SD	Mean	SD
pls	4.18	2.02	3.71	1.08
rf	4.10	2.21	4.41	1.40
lasso	3.99	1.83	1.41	0.72
rpart	3.70	1.94	6.49	0.85
glm	4.21	1.99	3.56	1.12
knn	4.06	2.22	5.96	1.08
cf	3.76	1.68	2.46	1.29

Notes: Mean and standard deviation of each learner’s rank by median-aggregated bias-corrected goodness-of-fit $\tilde{\Lambda}^{bc}$, computed across 10,000 Monte Carlo replications. Within each replication, learners are ranked from 1 (highest $\tilde{\Lambda}^{bc}$) to 7. Homogeneous DGP: $Y = \varepsilon$. Heterogeneous DGP: $Y = D \cdot \max(0, Z_1) + \varepsilon$. Sample size $n = 1,000$, $W^{inf} = W^{sel} = 100$ splits per replication.

selecting the top-performing learner based on the standard median-aggregated $\hat{\Lambda}(\ell)$. Size is held under the nominal level, and power is nearly 80 percent. If instead a researcher chose to select the top two learners as near-winners and make an *ad hoc* adjustment to the confidence interval bounds, the power drops nearly 16 percent with a Bonferroni correction, and 11.3 percent when reporting the union of confidence interval bounds. The MCS procedure, by contrast, achieves a rejection rate of 76.4 percent under the heterogeneous DGP—a loss of only 3.2 percentage points relative to naive top-1 selection. This is notable given that the MCS retains an average of 1.38 learners under the alternative, fewer than fixed top-2 selection but with comparable computational footprint and substantially better power. The key difference is adaptivity: the MCS selects $|\hat{\mathcal{M}}| = 1$ in 73 percent of replications, applying no correction at all, and expands to two or more learners only when the data cannot distinguish the best from its competitors. By contrast, the fixed top-2 rule always corrects for two learners regardless of whether the second-best learner is genuinely competitive. Under the homogeneous DGP, the MCS expands to a median of 5 learners with roughly uniform membership across all seven candidates, correctly reflecting that no learner outperforms any other when the CATE is constant.

Table 2 reports the range of median-aggregated $\hat{\beta}_2$ estimates across learners retained by the MCS. Under the homogeneous DGP, the MCS retains multiple learners in 87.5 percent of replications, and the average range of point estimates across the surviving set in these cases is 0.54. Since

Figure (1) Distribution of bias-corrected $\tilde{\Lambda}^{bc}$ by learner across Monte Carlo replications

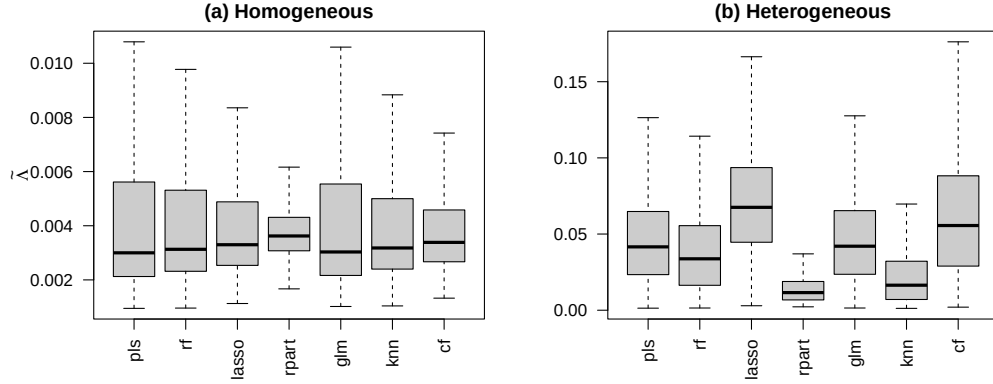


Table (2) Sensitivity of $\hat{\beta}_2$ to learner choice within the MCS

	Homogeneous	Heterogeneous
Fraction $ \hat{\mathcal{M}} > 1$	0.875	0.266
Mean range of $\hat{\beta}_2$ across $\hat{\mathcal{M}}$	0.470	0.166
Mean range when $ \hat{\mathcal{M}} > 1$	0.537	0.622
Fraction with range > 0.1	0.823	0.262
Fraction with range > 0.5	0.450	0.165

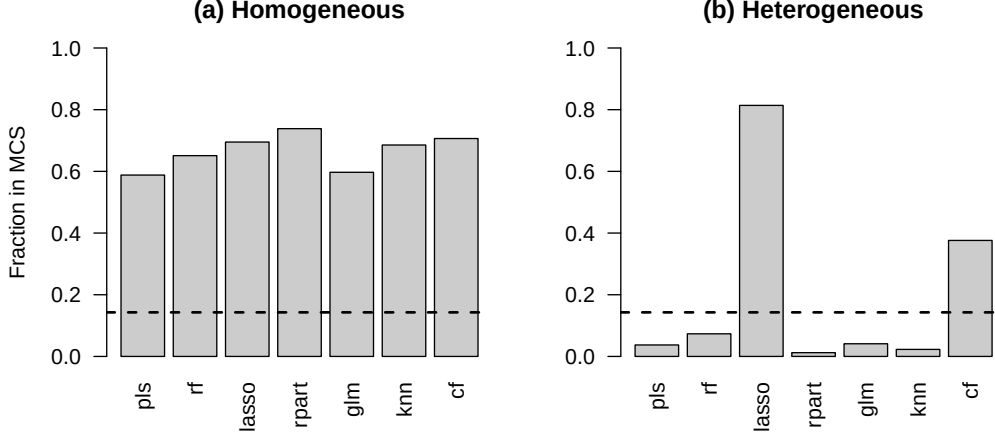
Notes: Range is computed as $\max_{\ell \in \hat{\mathcal{M}}} \tilde{\theta}_\ell - \min_{\ell \in \hat{\mathcal{M}}} \tilde{\theta}_\ell$, where $\tilde{\theta}_\ell$ is the median-aggregated $\hat{\beta}_2$ for learner ℓ . When $|\hat{\mathcal{M}}| = 1$, range is zero. Statistics computed across 10,000 Monte Carlo replications. MCS constructed at $\alpha_1 = 0.05$ over $W^{inf} = W^{sel} = 100$ splits with $B = 10,000$ bootstrap replications.

the true β_2 is zero, this variation is entirely driven by estimation noise, yet all of these estimates are equally defensible given the MCS. A researcher reporting only the naive top-1 estimate would present a single number without any indication that comparably performing learners disagree by over half a unit.¹⁵ Under the heterogeneous DGP, the MCS retains multiple learners less frequently (27 percent of replications), but when it does, the average range is 0.62. This variation reflects genuine differences in how learners approximate the CATE and has direct implications for applied work: two learners that the MCS deems statistically indistinguishable may produce meaningfully different estimates of the degree of treatment effect heterogeneity.

One might ask whether a simpler, threshold-based rule could achieve similar results—for example, defining the near-winner set as all learners within some fraction c of the maximum $\tilde{\Lambda}$. Such a rule requires the researcher to choose c , and the choice has direct consequences: too small and the set

¹⁵On average, the point estimate is 0.024.

Figure (2) Fraction of Monte Carlo replications in which each learner is included in the MCS



collapses to a singleton regardless of whether near-winners exist; too large and the set includes clearly inferior learners, unnecessarily widening confidence intervals. Moreover, the appropriate value of c depends on the dispersion of $\hat{\Lambda}$ across splits, which varies with the DGP, sample size, and learner set. The MCS replaces this subjective choice with a formal test calibrated at level α_1 . When the data strongly separate learners, the equivalence test rejects repeatedly and the set shrinks; when learners are indistinguishable, the test accepts early and the set remains large. This behavior is governed entirely by the sampling variability of the split-level performance values, not by a tuning parameter.¹⁶

6 Applications

This section applies the MCS procedure in two empirical settings that provide complementary illustrations of the learner-selection problem in GML. The first is Bryan et al. (2024), henceforth BKO, who use Generic Machine Learning to study heterogeneity in the returns to business loans in Egypt. BKO provide a setting in which the GML proxies reveal economically meaningful treatment-effect heterogeneity and learner choice matters for the resulting inference. The second is Hrozencik et al. (2024), who study a randomized social-comparison intervention designed to reduce agricultural

¹⁶Two additional experiments are underway. First, a coverage simulation targeting the population GATES contrast $\gamma_G - \gamma_1$ evaluated on groups defined by the true CATE, which is analytically tractable under the heterogeneous DGP and provides a well-defined target for empirical evaluation of Theorem 1. Second, a sensitivity analysis over the selection share ρ^{sel} to characterize the efficiency trade-off between the selection and inference stages, alongside a comparison to the aggregated variant that repeats the outer split and median-aggregates the resulting intervals.

Table (3) Size and Power for BLP, $n = 1000$, $W = 200$, $\alpha_1 = \alpha_2 = 0.05$

Method	Hom (size)	Het (power)	Mean $ \hat{\mathcal{M}} $	
			Hom	Het
Naive top-1, unadjusted	0.005	0.796	1	1
Naive top-2, BF(2)	0.002	0.683	2	2
Naive top-2, union	0.002	0.538	2	2
MCS + BF, best-in- $\hat{\mathcal{M}}$	0.004	0.764	4.67	1.38
MCS + union	0.004	0.755	4.67	1.38

Notes: Rejection rates of $H_0 : \beta_2 = 0$ across 10,000 Monte Carlo replications, $n = 1,000$, $W^{inf} = W^{sel} = 100$, $\alpha_1 = \alpha_2 = 0.05$. Homogeneous DGP: $Y = \varepsilon$ (size). Heterogeneous DGP: $Y = D \cdot \max(0, Z_1) + \varepsilon$ (power). Learner set: pls, rf, lasso, rpart, glm, knn, cf. Bias-corrected $\hat{\Lambda}^{bc}$ used for both MCS construction and naive selection. “Mean $|\hat{\mathcal{M}}|$ ” reports the mean size of the selected set under each procedure (fixed at 1 or 2 for naive rows). MCS constructed with $B = 10,000$ bootstrap replications.

groundwater use in Colorado and Kansas. In the GML specification considered below, the available learners provide little statistically detectable alignment with heterogeneous treatment effects. Taken together, the two applications illustrate two situations that applied researchers may encounter in practice: one in which flexible learners recover a strong heterogeneity signal, and another in which the available proxies provide little evidence that they have learned systematic variation in treatment effects.

The contrast is useful for evaluating the role of formal learner selection. The objective of the MCS is not to produce evidence of heterogeneity or to guarantee that a selected learner is a good approximation to the CATE. Rather, it quantifies uncertainty about relative learner performance given the information available in the data. In the BKO application, formal selection has direct consequences for the substantive inference reported from GML. The groundwater application provides a useful counterpoint: even when one learner performs better than the alternatives according to the GML goodness-of-fit criterion, subsequent inference need not indicate that the selected proxy contains meaningful information about treatment-effect heterogeneity.

6.1 BKO: Setting and Design

BKO study a randomized lending experiment among Egyptian microentrepreneurs. Borrowers in the treatment group were offered loans that were substantially larger than those offered to the control group, and the central empirical question is whether the gains from larger loans are heterogeneous across borrowers. To study this question, BKO combine rich psychometric and cognitive baseline data with the GML procedure of Chernozhukov et al. (2025a) to estimate heterogeneous treatment effects on monthly profits.

The experiment was implemented in partnership with the Alexandria Business Association, the largest microfinance institution in the Middle East by client count, under a program called Tamouh. Loan officers nominated existing clients whom they judged eligible for a substantially larger loan; these clients were then randomized into a treatment group offered loans roughly four times their previous loan and a control group offered loans roughly twice their previous loan. The intervention thus targets the “missing middle” of credit access in Egypt, where microfinance loan sizes typically cap below £E5,000 and bank lending begins above £E50,000. The outcome of primary interest is monthly business profits measured at follow-up. Treatment effect heterogeneity is studied along two distinct sets of borrower characteristics: a “standard” set comprising business performance and demographic measures of the type a lender would routinely collect, and a richer set augmenting the standard data with psychometric and cognitive measures, including assessments of optimism, risk aversion, and Raven’s matrices. The authors find statistically detectable heterogeneity only when the psychometric and cognitive measures are included, motivating their use of these data to construct the predicted treatment effects on which the GATES quartiles in their Table 4 are based. Accordingly, the application below uses the psychometric/cognitive specification, which is the input to the headline GATES results.

Their implementation uses four learners: elastic net, gradient boosting, neural net, and random forest. For each learner, the data are repeatedly split into auxiliary and main samples, first-stage proxies are learned in the auxiliary sample, and BLP and GATES parameters are estimated in the main sample. The final reported results are then obtained by aggregating across $W = 100$ splits. In the original analysis, BKO select the learner with the highest estimated goodness of fit and then apply a Bonferroni correction intended to account for learner search.

The corrigendum (Bryan et al., 2025) shows that the original code failed to retain the full collection of split-level results. Instead of storing all 100 splits, the relevant object was overwritten so that the reported results for Panel A of Table 4 and the BLP coefficients in Figure 2 reflected the final split rather than the median across splits. Once this error is corrected, the top-group GATES estimate remains positive and statistically significant, while the bottom-group estimate becomes smaller in magnitude and is no longer statistically different from zero. The corrected BLP coefficient also remains statistically significant. These revisions leave the qualitative evidence of heterogeneity intact, but they weaken the claim that a subset of borrowers is significantly harmed by larger loans.

6.2 Coverage Accounting in the Original Implementation

This application also highlights a separate issue concerning coverage. In the GML framework, median aggregation across splits requires an adjustment to the within-split confidence level. If one additionally wishes to account for selection across multiple learners, a second adjustment is needed

for multiplicity. These two steps are conceptually distinct and should be tracked separately.

BKO describe their procedure as applying a Bonferroni correction across four learners. However, the critical value used in the code corresponds to a within-split interval construction that delivers 95 percent coverage at the individual-learner level after median aggregation, not the 97.5 percent per-learner coverage required for an overall 90 percent Bonferroni adjustment across four learners. Put differently, the implemented procedure is consistent with the critical value needed when no four-learner penalty is imposed at the final inference stage, but it is not consistent with the coverage claim that would accompany a correctly implemented Bonferroni adjustment by four.

This distinction matters empirically. A correct Bonferroni adjustment across four learners at 90 percent overall coverage requires a larger critical value and therefore wider confidence intervals than those produced by the implemented code. The difference is not cosmetic. As shown below, it affects whether the top-group profit estimate remains statistically significant.

6.3 MCS Results

I apply the two-stage procedure to BKO’s experimental data using the corrected replication code. The outer split allocates $\rho^{\text{sel}} = 0.3$ of the sample to selection, with equal auxiliary/main splits on each side ($\rho_A^{\text{sel}} = \rho_A^{\text{inf}} = 0.5$). Within the selection sample, proxies are trained and split-level goodness-of-fit values are computed over $W^{\text{sel}} = 100$ auxiliary/main splits, and the reported inference is median-aggregated over $W^{\text{inf}} = 100$ splits of the inference sample. Following the logic developed above, the loss for learner ℓ in split w is taken to be the negative of a bias-corrected goodness-of-fit measure. In the baseline specification, I use the GATES-based criterion

$$\bar{\Lambda}(\ell) = \frac{1}{G} \sum_{k=1}^G \gamma_{k,\ell}^2,$$

together with the split-level bias correction

$$\bar{\Lambda}_w^{bc}(\ell) = \frac{1}{G} \sum_{k=1}^G (\hat{\gamma}_{k,\ell w}^2 - \hat{\text{se}}_{k,\ell w}^2),$$

and construct the MCS at level $\alpha_1 = 0.05$ using $B = 10,000$ bootstrap replications.

Panel A of Table 4 reports the resulting MCS. The selected set is a singleton consisting of random forest, and the remaining learners are eliminated immediately. This pattern is consistent with BKO’s original ranking of learners by goodness of fit. It is also stable across the alternative choices considered in the code, namely BLP-based versus GATES-based goodness-of-fit and raw versus bias-corrected loss measures. In this application, then, the data do not merely suggest that random forest is the best learner in sample. They suggest that the remaining learners are

sufficiently inferior that no additional Bonferroni adjustment across surviving learners is needed at the reporting stage. The selection cost is accounted for through the α_1 component of the two-stage coverage guarantee.

6.4 Inference

Panel B of Table 4 compares inference under three approaches: the procedure implemented in BKO’s code, a correctly implemented Bonferroni adjustment by four targeting 90 percent overall coverage, and the two-stage MCS procedure with overall coverage $1 - \alpha_1 - \alpha_2 = 0.90$, where $\alpha_1 = \alpha_2 = 0.05$. Because the MCS selects a singleton, the second-stage inference problem is reduced to ordinary median-aggregated inference for one learner on the inference sample. In consequence, the MCS does not pay a multiplicity penalty at the reporting stage. Three implications follow.

First, the evidence for heterogeneity itself is robust. The BLP heterogeneity parameter remains statistically different from zero under all approaches considered. Thus, regardless of how one handles multiplicity, the data continue to reject the null of no systematic treatment-effect heterogeneity.

Second, the bottom-group GATES estimate is not statistically significant once the coding error documented in the corrigendum is corrected. This conclusion is not driven by the choice between Bonferroni and MCS adjustments. Rather, it reflects the change in the underlying point estimate after the split-storage error is fixed. The corrected estimate is materially smaller in magnitude than the value reported in the original paper, and the resulting interval includes zero under every inferential procedure considered here.

Third, the top-group GATES estimate is precisely where the choice of multiplicity adjustment matters. Under the procedure implemented in BKO’s code, the top-group effect remains statistically significant. The same is true under the MCS procedure, because the learner-selection stage collapses to a singleton and no additional second-stage penalty is required. By contrast, under a correctly implemented Bonferroni adjustment by four at 90 percent overall coverage, the confidence interval widens enough that the lower bound crosses zero. Hence, if one were to insist on a four-learner Bonferroni correction at the final inference stage, the data would no longer support the conclusion that the top quartile of predicted beneficiaries experiences a statistically significant gain in profits.

This comparison clarifies the practical value of the MCS. The issue is not that the MCS mechanically restores significance. Rather, it restores power only when the data support the claim that the apparent search over learners is illusory in a statistical sense. In this application, the split-level performance comparisons indicate that random forest is the only learner that survives formal scrutiny. Once that is recognized, applying a Bonferroni penalty for learners that are not empirically competitive is unnecessarily conservative.

Finally, the GATES contrast between the top and bottom groups remains statistically significant under all approaches. Thus, even though the corrected data no longer support a statistically

significant negative effect for the bottom group in isolation, they continue to imply a meaningful spread between predicted high-return and low-return borrowers. The main substantive lesson of BKO therefore survives: larger loans generate substantial heterogeneity, and the economic value of improved targeting remains considerable.

Under the proper 90 percent multiplicity correction across the four learners considered by BKO, none of the four GATES coefficients is statistically distinguishable from zero, and the only surviving significant finding is the contrast $\gamma_4 - \gamma_1$. The MCS singleton avoids the multiplicity penalty entirely and recovers significance for the top group at the same 90% level, restoring the central economic finding that top-quartile borrowers gain substantially from larger loans. The bottom-group result remains insignificant under both procedures once the corrigendum is applied. Formal learner selection therefore changes which conclusion the data support: a robust positive return at the top, no detectable harm at the bottom, and a significant top-bottom contrast.

6.5 Groundwater Conservation: Setting and Design

Hrozencik et al. (2024) study a randomized field experiment testing whether social-comparison information can reduce groundwater use among commercial irrigators in Colorado and Kansas. Well owners and operators were randomized to receive an annual mailer comparing the groundwater use of selected wells with the distribution of use among nearby irrigators. Treatment assignment occurred at the owner/operator level and remained fixed throughout the experiment, while groundwater use was observed annually at the well level. The intervention reduced groundwater pumping by approximately 4 percent on average.

The authors also conduct a preregistered analysis of treatment-effect heterogeneity based on pre-intervention groundwater use. Their primary specification exploits the full 2011–2021 panel of well-level annual pumping and allows treatment effects to differ according to whether an owner/operator, or an individual well, had relatively high water use before the experiment. They find larger conservation responses among initially lower-use irrigators: groundwater use falls by approximately 6 percent for the lower-use groups, compared with roughly 1–2 percent for the corresponding higher-use groups. Thus, the original analysis provides evidence of heterogeneity along a specific, prespecified dimension of historical pumping behavior.

For the present application, I recast the experiment into an endpoint-style RCT suitable for the GML framework used throughout this paper. Because treatment was randomized at the owner/operator level, the GML observation is defined at that same level. For each owner/operator, I aggregate post-treatment groundwater use across the eligible wells and experimental years to form a single post-treatment outcome. Randomized treatment assignment, rather than realized mail receipt, is used as the treatment indicator, so the parameter of interest is an intent-to-treat effect. The transformation results in 3,247 total observations—one per owner/operator—1,631 of

which are treated and 1,616 are in the control group. The feature set is constructed entirely from pre-treatment information, including historical groundwater-use levels and patterns, irrigation intensity, operation scale, and other characteristics available before treatment assignment. The same four candidate learners and outer-split MCS procedure described above are then applied to this owner-level sample.

This transformation is important for interpreting the results. The analysis does not attempt to reproduce the panel heterogeneity specification of Hrozencik et al. (2024). Rather, it asks a different question: whether flexible machine-learning proxies constructed from available pre-treatment characteristics can detect systematic heterogeneity in the owner-level treatment effect. The distinction matters because failure to reject the GML null $\beta_2 = 0$ does not establish that the true CATE is homogeneous. Since

$$\beta_2 = \frac{\text{Cov}\{s_0(Z), S(Z)\}}{\text{Var}\{S(Z)\}},$$

a small or statistically insignificant β_2 may arise either because treatment effects are approximately homogeneous or because the learner simply fails to detect the heterogeneity in the underlying CATE.

This is precisely the pattern observed in the groundwater application. The original authors find heterogeneity under a specific prespecified moderator in their rich panel specification, while the GML application asks whether a collection of flexible learners can recover heterogeneous effects from the endpoint-style experimental design. The available proxies do not produce statistically detectable alignment with treatment-effect heterogeneity. Accordingly, the results should not be interpreted as evidence that the underlying CATE is homogeneous. Instead, they provide an example in which the available learners reveal little detectable heterogeneity, despite the existence of economically plausible and independently documented dimensions along which treatment responses may vary.

6.6 Groundwater Results

Table 5 reports the learner-selection and inference results for the groundwater application. The MCS reduces the four candidate learners to a singleton containing random forest. Thus, unlike a setting in which weak relative performance differences leave many learners in the confidence set, the selection sample is sufficiently informative to distinguish among the candidate learners. At the same time, the magnitudes of the goodness-of-fit measures provide little indication that any learner captures a strong heterogeneity signal. The bias-corrected criterion is small even for random forest, $\tilde{\Lambda}^{bc} = 0.00090$, and is negative for each of the three eliminated learners. Because the bias correction removes estimated sampling variation from the raw goodness-of-fit measure, these negative values should not be interpreted as negative population goodness of fit. Rather, they indicate that the estimated heterogeneity signal for these learners is sufficiently weak that the correction for estimation noise dominates it. The MCS therefore identifies a relatively superior

learner, but relative superiority does not imply strong performance in an absolute sense.

The independent inference sample reinforces this distinction. For the selected random forest, the median-aggregated BLP heterogeneity coefficient is $\tilde{\beta}_2 = -0.161$, with a 90 percent union confidence interval of $[-0.648, 0.343]$. Hence, the GML analysis provides no statistically detectable alignment between the selected proxy and heterogeneity in the true CATE. Importantly, this result does not imply that the underlying treatment effect is homogeneous. Hrozencik et al. (2024) detect heterogeneity along a prespecified dimension of pre-intervention water use using their richer panel specification; the present exercise asks instead whether flexible ML proxies can recover systematic heterogeneity from the owner-level endpoint design. In this setting they do not. The application therefore illustrates an important distinction: the MCS can identify the best-performing learner among the available candidates while subsequent GML inference still reveals that even the selected learner contains little detectable information about treatment-effect heterogeneity.

7 Conclusion

This paper proposes applying the model confidence set of Hansen et al. (2011) to the learner selection problem in Generic Machine Learning. The procedure provides a data-driven, formally justified determination of which learners are statistically indistinguishable under a user-specified loss criterion, replacing the ad hoc selection that currently characterizes applied practice. As a default loss, I propose a bias-corrected version of the GML goodness-of-fit functional that removes the within-split component of the noise inflation in the squared BLP coefficient, reducing the tendency of the raw criterion to mechanically reward learners with unstable proxies. The resulting near-winner set feeds naturally into the multiplicity corrections already recommended by Chernozhukov et al. (2025a), with the key advantage that the size of the correction adapts to the relative performance of the candidate learners. The two-stage coverage guarantee at level $1 - \alpha_1 - \alpha_2$ is delivered by an initial sample split that reserves the observations used for MCS selection, so that the reported inference uses only observations that never entered the selection stage.

Three findings emerge from the Monte Carlo analysis. First, the two-stage procedure controls size at the nominal level across all specifications considered. Second, the power cost relative to naive selection is modest when the MCS is able to eliminate inferior learners, and can be particularly small when a clear winner produces a singleton set. Third, when multiple learners remain empirically credible, the MCS reveals meaningful variation in their point estimates, providing applied researchers with a diagnostic of learner uncertainty that is unavailable under standard reporting practices.

The two empirical applications illustrate different roles for formal learner selection. In the application to Bryan et al. (2024), the GML analysis produces clear evidence of treatment-effect heterogeneity and the MCS selects a singleton learner. The procedure therefore implies that no

additional second-stage correction across multiple surviving learners is required; the cost of learner selection is instead borne by the α_1 component of the two-stage guarantee. This sharpens the interpretation of the corrected results of Bryan et al. (2025). Under a proper multiplicity correction across all four initially considered learners, the top-quartile profit effect is no longer statistically significant at the conventional 90% level. Formal selection shows, however, that the data do not support treating all four learners as empirically competitive. Once the inferior learners are removed, the significant positive effect for the top quartile is recovered while the corrected bottom-group effect remains statistically indistinguishable from zero.

The groundwater application provides a useful counterpoint. Hrozencik et al. (2024) find heterogeneous responses to a social-comparison intervention along a prespecified measure of historical groundwater use in their rich panel specification. When the experiment is recast into the owner-level endpoint design used for GML here, however, the candidate ML proxies provide little statistically detectable alignment with heterogeneous treatment effects. The MCS nevertheless selects random forest as the relatively best-performing learner. This distinction is important: identifying the best learner among a candidate set is not equivalent to establishing that the selected learner is a strong proxy for the CATE. In the independent inference sample, the selected learner’s BLP heterogeneity coefficient remains statistically indistinguishable from zero. This result should not be interpreted as evidence that the true CATE is homogeneous. Rather, it illustrates a setting in which the available flexible learners fail to recover detectable systematic heterogeneity under the chosen GML specification. Together, the two applications show that formal learner selection can sharpen inference when informative proxies exist without manufacturing evidence of heterogeneity when they do not.

Several directions for future work remain. The efficiency trade-off implicit in the outer split—between the observations available for selection and those available for the reported inference—warrants systematic study, as the appropriate proportions ρ^{sel} , ρ_A^{sel} , and ρ_A^{inf} likely depend on the DGP and candidate learner set. A related question is the aggregation of results across repeated outer splits, which restores full-sample symmetry at the cost of a larger error budget. The MCS framework may also be compatible with the cross-fitting approach of Imai and Li (2025) and the sequential framework of Wager (2025), though formal integration requires care: the i.i.d. split structure that underpins the nonparametric bootstrap in the present setting may not hold under alternative splitting schemes, potentially necessitating a different resampling procedure. More broadly, the idea of using MCS to select among competing machine learning methods before conducting inference may extend to other causal frameworks that make use of sample splitting, such as double/debiased machine learning (Chernozhukov et al., 2018), but verifying the regularity conditions and identifying an appropriate loss criterion in each setting remains an open question.

Table (4) Treatment Impacts on Firm Profits: Learner Selection and Inference

Panel A: Learner Rankings and MCS Membership

Learner	$\tilde{\Lambda}$	$\tilde{\Lambda}^{bc}$	MCS p -value	In $\hat{\mathcal{M}}_{0.05}$?
Random forest	20,462,959	13,505,075	1.000	Yes
Gradient boosting	13,817,836	7,200,878	0.000	No
Neural net	7,639,281	901,266	0.000	No
Elastic net	8,151,449	433,251	0.000	No

Panel B: Median-Aggregated Inference under Three Approaches

Parameter	Estimate	Bonferroni		MCS	
		BKO code	$\times 4$ (90%)	Union	BF
$\hat{\beta}_2$	0.693	[0.108, 1.370]*	[0.037, 1.440]*	[0.108, 1.370]*	[0.108, 1.370]*
$\hat{\gamma}_1$	-4,556	[-10,194; 1,188]	[-10,881; 1,774]	[-10,194; 1,188]	[-10,194; 1,188]
$\hat{\gamma}_2$	273	[-4,530; 4,740]	[-5,060; 5,270]	[-4,530; 4,740]	[-4,530; 4,740]
$\hat{\gamma}_3$	1,985	[-2,752; 6,985]	[-3,263; 7,524]	[-2,752; 6,985]	[-2,752; 6,985]
$\hat{\gamma}_4$	6,708	[328; 12,788]*	[-390; 13,620]	[328; 12,788]*	[328; 12,788]*
$\hat{\gamma}_4 - \hat{\gamma}_1$	11,065	[2,809; 19,519]*	[2,052; 20,700]*	[2,809; 19,519]*	[2,809; 19,519]*

Notes: Panel A reports the median GATES-based goodness-of-fit $\tilde{\Lambda} = \text{med}_w G^{-1} \sum_k \hat{\gamma}_{k,\ell w}^2$ and its bias-corrected analog $\tilde{\Lambda}^{bc} = \text{med}_w G^{-1} \sum_k (\hat{\gamma}_{k,\ell w}^2 - \hat{s}_{k,\ell w}^2)$ for each learner. The MCS is constructed at level $\alpha_1 = 0.05$ using 10,000 bootstrap replications and $W^{inf} = W^{sel} = 100$ splits. Panel B reports point estimates (median across splits) and confidence intervals for the BLP heterogeneity parameter β_2 and GATES parameters γ_k ($k = 1, \dots, 4$ quartiles of predicted treatment effects on profits). “BKO code” reproduces the critical value used by Bryan et al. (2024) ($z = 2.241$); under the Bonferroni $\times 4$ correction the authors state they apply, this provides 80% overall coverage. “ $\times 4$ (90%)” applies the correct Bonferroni $\times 4$ critical value for 90% coverage ($z = 2.498$). “MCS Union” and “MCS BF” report inference over the model confidence set at 90% overall coverage ($1 - \alpha_1 - \alpha_2 = 0.90$). Since $|\hat{\mathcal{M}}_{0.05}| = 1$, both MCS columns use $z = 2.241$ with no multiplicity adjustment, and the MCS columns coincide with BKO code. * denotes that the interval excludes zero.

Table (5) Learner Selection and Heterogeneity in the Groundwater Application

Learner	Selection stage				Inference stage	
	$\tilde{\Lambda}^{bc}$	Rank	MCS p -value	In $\hat{\mathcal{M}}_{0.05}$?	$\tilde{\beta}_2$	Union CI
Lasso	-0.00058	2	0.0236	No	-0.342	—
Random forest	0.00090	1	1.0000	Yes	-0.161	[-0.648, 0.343]
XGBoost	-0.00174	4	0.0236	No	-0.097	—
Causal forest	-0.00098	3	0.0012	No	-0.187	—

Notes: Selection based on $\tilde{\Lambda}^{bc}$. The MCS is constructed at $\alpha_1 = 0.5$ using 10,000 bootstrap draws and $W^{inf} = W^{sel} = 100$ splits. $\alpha_2 = 0.05$, yielding union CI coverage of the point estimate after selection of 90%.

A Additional Results

Table (6) Mean $\tilde{\Lambda}$ and bias-corrected $\tilde{\Lambda}^{bc}$ by learner

	Homogeneous			Heterogeneous		
	$\tilde{\Lambda}$	$\tilde{\Lambda}^{bc}$	Diff.	$\tilde{\Lambda}$	$\tilde{\Lambda}^{bc}$	Diff.
pls	0.00502	-0.00308	0.00811	0.04751	0.03898	0.00853
rf	0.00474	-0.00338	0.00812	0.03897	0.03041	0.00856
lasso	0.00449	-0.00362	0.00811	0.07174	0.06320	0.00853
rpart	0.00378	-0.00434	0.00812	0.01416	0.00559	0.00856
glm	0.00499	-0.00311	0.00810	0.04795	0.03941	0.00854
knn	0.00469	-0.00343	0.00812	0.02217	0.01361	0.00856
cf	0.00424	-0.00381	0.00805	0.06255	0.05401	0.00854
Spearman ρ	0.90650			0.98940		

Notes: Mean median-aggregated raw $\tilde{\Lambda}$ and bias-corrected $\tilde{\Lambda}^{bc}$ by learner across 10,000 Monte Carlo replications. “Diff.” is $\tilde{\Lambda} - \tilde{\Lambda}^{bc}$. Spearman’s ρ is computed between the raw and bias-corrected rankings within each replication and averaged across replications. DGPs and configuration as in Table 1.

B Proofs

The proof of Proposition 1 maps the GML setting to the MCS framework of Hansen et al. (2011)—now abbreviating as HLN—and invokes their main coverage result. GML fits the framework naturally, so the proof relies heavily on the results of HLN. In what follows, “coherency” refers to a test-elimination pair for which the probability of simultaneously keeping and rejecting a superior learner is bounded by the null rejection probability (see HLN Definition 3). The relative loss variables $\{d_{ij,w}\}_{w=1}^{W^{\text{sel}}}$ are i.i.d. across w conditional on $\mathcal{D}^{\text{sel}}$ by Assumption 1, and Assumption 2 establishes finite losses (hence, by finiteness of the split collection, finite moments of all orders) and nondegenerate relative-to-average losses conditional on $\mathcal{D}^{\text{sel}}$. The arguments below therefore deliver HLN Theorem 1 applied to the conditional sequence, yielding the bound stated in Proposition 1.

Proof of Proposition 1. The relative loss variables $\{d_{ij,w}\}$ must satisfy the regularity conditions of HLN Assumption 2 which requires strict stationarity, α -mixing of order $-r/(r-2)$ for some $r > 2$, finite $r + \gamma$ moments $\gamma > 0$, and positive variance. The GML relative losses $\{d_{ij,w}\}_{w=1}^{W^{\text{sel}}}$ —as defined in (7)—are i.i.d. across w under Assumption 1, so stationarity and α -mixing of any order follow immediately in this special case. Conditional on $\mathcal{D}^{\text{sel}}$, the split distribution $U(\mathcal{A})$ has finite support, so Assumption 2 (i) implies that the losses, and hence the relative losses, have finite moments of all orders, delivering the moment condition. Finally, Assumption 2 (ii) ensures positive variance of the relative-to-average losses entering the test statistics.

The elimination rule (13) and the tests based on $T_{\max, \mathcal{M}}$ of (12) are the same formulation as HLN. Thus, the test and elimination rule can be said to have “coherency” (HLN Definition 3) by an appeal to HLN Proposition 1. Under HLN Assumption 2 (verified above) HLN Theorem 4 establishes asymptotic size, power, and elimination behavior for the $T_{\max, \mathcal{M}}$ -based test, which delivers the requisite conditions of HLN Assumption 1. Invoking HLN Theorem 1, applied to the loss process generated by repeated splits of $\mathcal{D}^{\text{sel}}$ conditional on $\mathcal{D}^{\text{sel}}$, delivers the result of Proposition 1. $\square$

Proof of Theorem 1. Define the events

$$\begin{aligned} A &= \{\ell^* \in \hat{\mathcal{M}}_{1-\alpha_1}\}, \\ B &= \{\theta_{\ell^*} \in \tilde{I}_{\theta_{\ell^*}}(\alpha_2)\}, \\ C &= \{\theta_{\ell^*} \in \tilde{I}^U(\alpha_2)\}. \end{aligned}$$

By construction of the union interval, $\tilde{I}_{\theta_{\ell^*}}(\alpha_2) \subseteq \tilde{I}^U(\alpha_2)$ whenever $\ell^* \in \hat{\mathcal{M}}_{1-\alpha_1}$, so $A \cap B \subseteq C$. Taking complements and applying De Morgan’s law, $C^c \subseteq A^c \cup B^c$. By monotonicity of probability and Boole’s inequality,

$$\Pr(C^c) \leq \Pr(A^c) + \Pr(B^c).$$

The outer split of Section 3.1 plays the essential role in bounding $\Pr(B^c)$. Let U_0 denote the randomization variable generating the outer partition, and let U^{sel} and U^{inf} denote the randomization variables generating the selection-stage and inference-stage splits; all three are drawn independently of the data and of one another. Conditional on U_0 , the samples $\mathcal{D}^{\text{sel}}$ and $\mathcal{D}^{\text{inf}}$ are disjoint collections of i.i.d. observations and are therefore independent, so the sigma-algebras

$$\mathcal{G}^{\text{sel}} = \sigma(\mathcal{D}^{\text{sel}}, U^{\text{sel}}), \quad \mathcal{G}^{\text{inf}} = \sigma(\mathcal{D}^{\text{inf}}, U^{\text{inf}})$$

are conditionally independent given U_0 . Because this conditional joint distribution does not depend on the realization of U_0 , integrating over the distribution of U_0 yields the same factorizations unconditionally; in particular, $\mathcal{G}^{\text{sel}}$ and $\mathcal{G}^{\text{inf}}$ are independent. Every selection-stage object is $\mathcal{G}^{\text{sel}}$ -measurable: the losses $\{L_{\ell w}\}$, the model confidence set $\hat{\mathcal{M}}_{1-\alpha_1}$, the conditional superior set $\mathcal{M}^*(\mathcal{D}^{\text{sel}})$ (a deterministic functional of $\mathcal{D}^{\text{sel}}$), and therefore ℓ^* under any deterministic tie-breaking rule. Every inference-stage object is $\mathcal{G}^{\text{inf}}$ -measurable: the realized split collection $\mathcal{A}^{\text{inf}}$, the per-split estimates $\{\hat{\theta}_{\ell w}, \hat{\text{se}}_{\ell w}\}$, the targets $\theta_\ell = \text{Med}[\theta_{\ell, A} \mid \mathcal{D}^{\text{inf}}, \mathcal{A}^{\text{inf}}]$, and the intervals $\tilde{I}_{\theta_\ell}(\alpha_2)$.

For each fixed $\ell \in \mathcal{L}$, CDDF Theorem 4.3(i) applied to the inference sample—an i.i.d. sample of size $\rho^{\text{inf}} N$ with auxiliary/main proportions $(\rho_A^{\text{inf}}, 1 - \rho_A^{\text{inf}})$ and the realized split collection $\mathcal{A}^{\text{inf}}$ —delivers the coverage guarantee (16),

$$\Pr(\theta_\ell \in \tilde{I}_{\theta_\ell}(\alpha_2)) \geq 1 - \alpha_2 - o_n(1),$$

where n indexes the inference-stage main-sample size $(1 - \rho_A^{\text{inf}})\rho^{\text{inf}} N$, the guarantee holds at the realized W^{inf} , and the $o_n(1)$ term does not depend on W^{sel} . The bound may be taken uniformly over $\ell \in \mathcal{L}$: since $\mathcal{L}$ is finite, the maximum of finitely many $o_n(1)$ sequences is itself $o_n(1)$. Since $\{\ell^* = \ell\} \in \mathcal{G}^{\text{sel}}$ and $\{\theta_\ell \in \tilde{I}_{\theta_\ell}(\alpha_2)\} \in \mathcal{G}^{\text{inf}}$ are independent events,

$$\Pr(B) = \sum_{\ell \in \mathcal{L}} \Pr(\ell^* = \ell, \theta_\ell \in \tilde{I}_{\theta_\ell}(\alpha_2)) = \sum_{\ell \in \mathcal{L}} \Pr(\ell^* = \ell) \Pr(\theta_\ell \in \tilde{I}_{\theta_\ell}(\alpha_2)) \geq 1 - \alpha_2 - o_n(1),$$

where the final inequality uses $\sum_{\ell} \Pr(\ell^* = \ell) = 1$. Hence $\Pr(B^c) \leq \alpha_2 + o_n(1)$. Note that no property of the selection rule is used beyond $\mathcal{G}^{\text{sel}}$ -measurability; the same bound holds for any learner index determined by the selection stage.

Since $\ell^* \in \mathcal{M}^*(\mathcal{D}^{\text{sel}})$, the event $\{\mathcal{M}^*(\mathcal{D}^{\text{sel}}) \subseteq \hat{\mathcal{M}}_{1-\alpha_1}\}$ is contained in A , so $\Pr(A^c) \leq \Pr(\mathcal{M}^*(\mathcal{D}^{\text{sel}}) \not\subseteq \hat{\mathcal{M}}_{1-\alpha_1})$. By Proposition 1(i), $\limsup_{W^{\text{sel}} \rightarrow \infty} \Pr(A^c \mid \mathcal{D}^{\text{sel}}) \leq \alpha_1$ almost surely. Taking expectations over $\mathcal{D}^{\text{sel}}$ and applying the reverse Fatou lemma (the sequence is bounded by one) yields $\limsup_{W^{\text{sel}} \rightarrow \infty} \Pr(A^c) \leq \alpha_1$. Combining, for each fixed n ,

$$\limsup_{W^{\text{sel}} \rightarrow \infty} \Pr(C^c) \leq \alpha_1 + \alpha_2 + o_n(1).$$

Since $\Pr(C) = 1 - \Pr(C^c)$ and $\liminf_W (1 - x_W) = 1 - \limsup_W x_W$ for any real sequence x_W ,

$$\liminf_{W^{\text{sel}} \rightarrow \infty} \Pr(C) \geq 1 - \alpha_1 - \alpha_2 - o_n(1).$$

Taking $n \rightarrow \infty$ yields the result. □

C Further Discussion on Loss Design

Proposition 1 guarantees that the MCS contains the conditionally superior learners regardless of which loss function is used, provided Assumptions 1 and 2 hold. However, the *identity* of the conditionally superior set $\mathcal{M}^*(\mathcal{D})$ depends on the loss. A poorly designed criterion could rank learners in an unintended way, producing a valid MCS for a meaningless ranking. This appendix section examines the loss-design properties of the bias-corrected criterion in (6), identifies the components of noise inflation that the correction does and does not address, and clarifies which properties of the resulting ranking are formally established and which are conditions on the loss design that lie outside the scope of the MCS validity guarantee.

Throughout this subsection, let $n = N/2$ denote the size of each subsample in a GML split, and let all expectations and variances be taken over the split randomness conditional on $\mathcal{D}$, suppressing the conditioning for readability.

Conditional performance

Define the conditional goodness-of-fit of learner ℓ as

$$\Lambda^*(\ell, \mathcal{D}) = \left(\mathbb{E} \left[\hat{\beta}_{2,\ell w} \right] \right)^2 \cdot \mathbb{E} \left[\hat{V}_{\ell w} \right], \quad (19)$$

where the expectations are over the random split w . The first factor is the squared average BLP coefficient across all admissible splits of $\mathcal{D}$, and the second is the average proxy variance. Both are properties of the realized dataset and the learner, and $\Lambda^*(\ell, \mathcal{D})$ is in principle computable by enumerating all admissible splits. A learner with a larger Λ^* produces, on average across splits, a proxy that explains more of the variation in the CATE as estimated from this dataset.

The loss $L_{\ell w}$ is well-designed if its conditional expectation preserves the ranking induced by Λ^* : that is,

$$\Lambda^*(i, \mathcal{D}) > \Lambda^*(j, \mathcal{D}) \implies \mathbb{E}[d_{ij,w}] < 0. \quad (20)$$

When (20) holds, the superior set $\mathcal{M}^*(\mathcal{D})$ defined by the loss coincides with the set of learners maximizing $\Lambda^*(\cdot, \mathcal{D})$, and the MCS procedure identifies the best-performing learners under a meaningful criterion.

The raw criterion inflates noisy learners

Consider first the uncorrected loss $L_{\ell w}^{\text{raw}} = -\hat{\beta}_{2,\ell w}^2 \hat{V}_{\ell w}$. To see why this may not satisfy (20), set aside $\hat{V}_{\ell w}$ momentarily and examine the squared coefficient. By the variance decomposition,

$$\mathbb{E} \left[\hat{\beta}_{2,\ell w}^2 \right] = \left(\mathbb{E} \left[\hat{\beta}_{2,\ell w} \right] \right)^2 + \text{Var} \left(\hat{\beta}_{2,\ell w} \right). \quad (21)$$

The first term is the squared signal; the second is a noise inflation term. Two learners with the same average signal but different across-split variances will be ranked differently by $\hat{\beta}_{2,\ell w}^2$, with the noisier learner appearing superior. Since the variance term is learner-specific—depending on how sensitive the learner’s predictions are to which observations land in the auxiliary versus main sample—this inflation can flip the ranking relative to Λ^* .

The bias correction

The bias-corrected loss replaces $\hat{\beta}_{2,\ell w}^2$ with $\hat{\beta}_{2,\ell w}^2 - \hat{\text{se}}_{\ell w}^2$, where $\hat{\text{se}}_{\ell w}$ is the OLS standard error from the BLP regression in split w . Under standard conditions on the design, $\hat{\text{se}}_{\ell w}^2$ consistently estimates the within-split sampling variance of $\hat{\beta}_{2,\ell w}$ holding the learned proxy $\hat{S}_\ell(Z)$ fixed:

$$\mathbb{E}[\hat{\text{se}}_{\ell w}^2 \mid \mathcal{D}_{A,w}, \mathcal{D}] \approx \text{Var}(\hat{\beta}_{2,\ell w} \mid \mathcal{D}_{A,w}, \mathcal{D}). \quad (22)$$

By the law of total variance,

$$\text{Var}(\hat{\beta}_{2,\ell w} \mid \mathcal{D}) = \underbrace{\mathbb{E}[\text{Var}(\hat{\beta}_{2,\ell w} \mid \mathcal{D}_{A,w}, \mathcal{D}) \mid \mathcal{D}]}_{\text{within-split component}} + \underbrace{\text{Var}(\mathbb{E}[\hat{\beta}_{2,\ell w} \mid \mathcal{D}_{A,w}, \mathcal{D}] \mid \mathcal{D})}_{\text{auxiliary-sample component}}. \quad (23)$$

Combining (23) with the variance identity $\mathbb{E}[\hat{\beta}_{2,\ell w}^2 \mid \mathcal{D}] = (\mathbb{E}[\hat{\beta}_{2,\ell w} \mid \mathcal{D}])^2 + \text{Var}(\hat{\beta}_{2,\ell w} \mid \mathcal{D})$ in (21) yields

$$\mathbb{E}[\hat{\beta}_{2,\ell w}^2 - \hat{\text{se}}_{\ell w}^2 \mid \mathcal{D}] = \underbrace{\left(\mathbb{E}[\hat{\beta}_{2,\ell w} \mid \mathcal{D}]\right)^2}_{\text{auxiliary-sample residual}} + \underbrace{\text{Var}(\mathbb{E}[\hat{\beta}_{2,\ell w} \mid \mathcal{D}_{A,w}, \mathcal{D}] \mid \mathcal{D})}_{\text{auxiliary-sample residual}}. \quad (24)$$

The correction therefore removes the within-split component exactly in expectation, but leaves a residual reflecting variation across auxiliary samples in the population BLP coefficient associated with the realized proxy. The size of this residual depends on how stable the learner’s proxy is across draws of $\mathcal{D}_{A,w}$. The GML framework deliberately does not assume the proxy is consistent for $s_0(Z)$ (CDDF, p. 1147), although CDDF Theorem 5.2 provides convergence to a best-in-class approximation $s_\bullet(Z)$ for their causal learners (A) and (B) under Rademacher complexity conditions. For learners outside that class, the magnitude of the residual is not characterized by general theory.

The MCS coverage guarantee in Proposition 1 does not depend on whether the auxiliary-sample residual is small. The procedure is valid for whatever ranking the loss induces, and a researcher who prefers the raw criterion $-\hat{\Lambda}_w(\ell)$, or any other admissible loss satisfying Assumptions 1–2, may use it. The substantive question of whether the bias-corrected loss preserves the ranking induced by $\Lambda^*(\ell, \mathcal{D})$ is treated separately below.

The cross-term

The full bias-corrected loss is the product $\hat{\Lambda}_w^{bc}(\ell) = (\hat{\beta}_{2,\ell w}^2 - \hat{s}_{\ell w}^2) \hat{V}_{\ell w}$. Decomposing the product expectation,

$$\mathbb{E}[\hat{\Lambda}_w^{bc}(\ell) | \mathcal{D}] = \mathbb{E}[\hat{\beta}_{2,\ell w}^2 - \hat{s}_{\ell w}^2 | \mathcal{D}] \cdot \mathbb{E}[\hat{V}_{\ell w} | \mathcal{D}] + \text{Cov}(\hat{\beta}_{2,\ell w}^2 - \hat{s}_{\ell w}^2, \hat{V}_{\ell w} | \mathcal{D}). \quad (25)$$

The first product is handled by (24): it equals $\Lambda^*(\ell, \mathcal{D})$ plus the auxiliary-sample residual identified above. The remaining task is to bound the cross-term covariance.

By the law of total covariance,

$$\text{Cov}(\hat{\beta}_{2,\ell w}^2 - \hat{s}_{\ell w}^2, \hat{V}_{\ell w} | \mathcal{D}) = \underbrace{\mathbb{E}[\text{Cov}(\hat{\beta}_{2,\ell w}^2 - \hat{s}_{\ell w}^2, \hat{V}_{\ell w} | \mathcal{D}_{A,w}, \mathcal{D}) | \mathcal{D}]}_{\text{within-auxiliary component}} \quad (26)$$

$$+ \underbrace{\text{Cov}(\mathbb{E}[\hat{\beta}_{2,\ell w}^2 - \hat{s}_{\ell w}^2 | \mathcal{D}_{A,w}, \mathcal{D}], \mathbb{E}[\hat{V}_{\ell w} | \mathcal{D}_{A,w}, \mathcal{D}] | \mathcal{D})}_{\text{across-auxiliary component}}. \quad (27)$$

The within-auxiliary component admits a sharp rate. Conditional on $\mathcal{D}_{A,w}$, the proxy $\hat{S}_\ell(Z)$ is a fixed function and $\hat{V}_{\ell w}$ is the sample variance of $\hat{S}_\ell(Z)$ over the n i.i.d. main-sample observations, so $\text{Var}(\hat{V}_{\ell w} | \mathcal{D}_{A,w}, \mathcal{D}) = O(n^{-1})$. Under Assumption 2(i), $\text{Var}(\hat{\beta}_{2,\ell w}^2 - \hat{s}_{\ell w}^2 | \mathcal{D}_{A,w}, \mathcal{D}) = O(1)$. By the Cauchy–Schwarz inequality,

$$\left| \text{Cov}(\hat{\beta}_{2,\ell w}^2 - \hat{s}_{\ell w}^2, \hat{V}_{\ell w} | \mathcal{D}_{A,w}, \mathcal{D}) \right| \leq O(n^{-1/2}),$$

uniformly across learners and across realizations of $\mathcal{D}_{A,w}$. Iterating the expectation, the within-auxiliary component of (26) is $O(n^{-1/2})$.

The across-auxiliary component does not generally shrink in n . It captures how $\mathbb{E}[\hat{\beta}_{2,\ell w}^2 - \hat{s}_{\ell w}^2 | \mathcal{D}_{A,w}, \mathcal{D}]$ (the squared population BLP coefficient associated with the realized proxy, modulo the auxiliary-sample residual already discussed) co-varies across draws of $\mathcal{D}_{A,w}$ with $\mathbb{E}[\hat{V}_{\ell w} | \mathcal{D}_{A,w}, \mathcal{D}]$ (the population variance of the realized proxy). Both quantities depend on $\mathcal{D}_{A,w}$ only through the learned proxy, so this component reflects exactly the auxiliary-sample variation in proxy quality and proxy variance identified in the previous subsection. Bounding it without invoking proxy stability assumptions—which are at odds with the agnostic spirit of GML—is not possible in general.

Combining the two components,

$$\mathbb{E}[\hat{\Lambda}_w^{bc}(\ell) | \mathcal{D}] = \Lambda^*(\ell, \mathcal{D}) + R_{\text{aux}}(\ell, \mathcal{D}) + O(n^{-1/2}), \quad (28)$$

where $R_{\text{aux}}(\ell, \mathcal{D})$ collects the auxiliary-sample residual from (24) and the across-auxiliary cross-term from (26), both of which depend on the learner’s proxy stability across $\mathcal{D}_{A,w}$. The $O(n^{-1/2})$

term is uniform across learners.

Since the MCS test statistics in (11)–(12) are sample averages of $d_{ij,w}$, which by Assumptions 1–2 converge in probability to $\mathbb{E}[d_{ij,w} \mid \mathcal{D}]$, ranking preservation by the bias-corrected loss requires

$$|R_{\text{aux}}(i, \mathcal{D}) - R_{\text{aux}}(j, \mathcal{D})| < |\Lambda^*(i, \mathcal{D}) - \Lambda^*(j, \mathcal{D})| \quad (29)$$

for all $i, j \in \mathcal{L}$ with $\Lambda^*(i, \mathcal{D}) \neq \Lambda^*(j, \mathcal{D})$, once n is large enough that the uniform $O(n^{-1/2})$ term is negligible.

Whether (29) holds in any given application is a separate question from MCS validity and beyond the scope of this paper. This is an area for further research. Direct verification would require estimating $\Lambda^*(\ell, \mathcal{D})$ for each learner via a high-resolution Monte Carlo over the split distribution $U(\mathcal{A})$ on a fixed realized dataset, which is a computationally-heavy task. I emphasize that ranking preservation against Λ^* is a property of the loss design, not of the MCS procedure: the coverage guarantee of Proposition 1 holds for any loss satisfying Assumptions 1–2, and the MCS returns a valid confidence set for the ranking that the chosen loss induces. The simulations in Section 5 document size control, power, and the membership behavior of the MCS in two specific designs; they are consistent with the bias-corrected loss producing a substantively useful ranking in those configurations, but they do not constitute a formal verification of (29).

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